

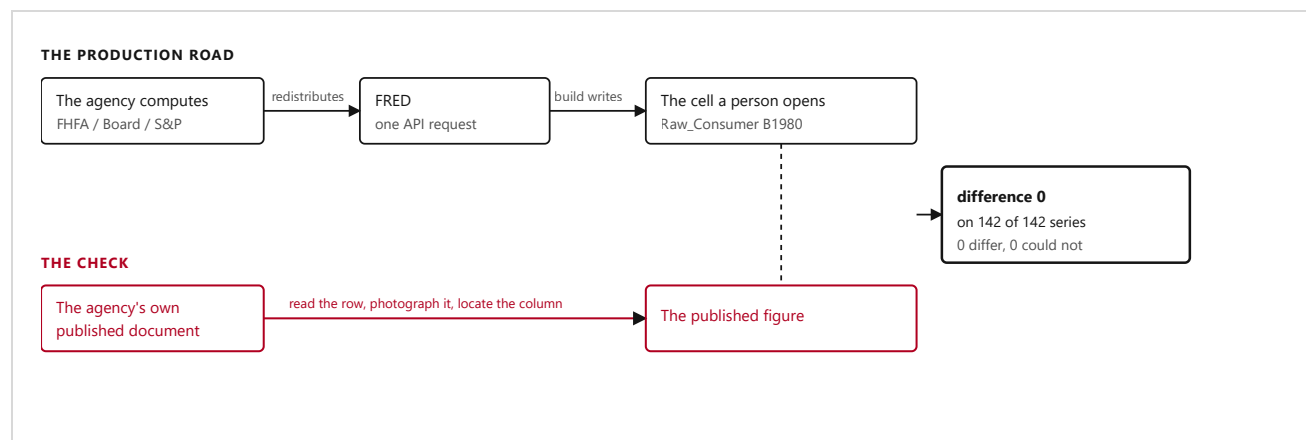
Tie-out: every number in the FRED credit monitor

142 series, each proved against the agency that computes it — not against FRED, which only redistributes them. 5 September 2026.

142 of 142 tied. 0 differ. 0 could not.

Every figure on the left of every comparison below was read out of the shipped workbook, `FRED_Credit_Risk_Dashboard.xlsm`, and every figure on the right came off a document published by the agency that computed it. The tie-out found four defects, one of which was shipping a state with no data in it, and all four are in “What this found” below.

How the two roads meet



Only the lower road touches a document nobody here controls. That is what makes this evidence rather than a second opinion from the same source: FRED is the redistributor, and every figure was checked against the agency that computed it.

One figure, traced all the way

The roster covers all 142. This is one of them followed link by link, so a reader can see the shape of the check before trusting the count.

1 • The figure

Total consumer credit outstanding, seasonally adjusted, at June 2026. In the workbook it is on the `Raw_Consumer` tab. Open the file, go to that tab and that cell, and this is what is there:

	A	B	C
1978	TOTALSL	Total Consumer C	(<i>freq=monthly; units=billions \$; transform=yoy_pct; vintage</i>
1979	date	value	
1980	2026-06-01	5166907.71	
1981	2026-05-01	5152735.06	
1982	2026-04-01	5153816.46	

FRED_Credit_Risk_Dashboard.xlsm, Raw_Consumer. Excel's own row numbers and column letters are printed, so the reference can be checked rather than taken on trust.

2 • The call that produced it

```
curl "https://api.stlouisfed.org/fred/series/observations?
series_id=TOTALSL&api_key=$FRED_API_KEY&file_type=json"
```

That is the build's request. It is *not* the check: FRED redistributes this series, so asking FRED whether FRED is right is the mirror. The check is link 4.

3 • What happens in between

Nothing. The series is a level and the workbook stores it as landed, in millions of dollars, newest observation first. There is no transform, no rescaling and no unit conversion between the response and the cell — which is worth saying out loud, because “then the system computes it” is exactly the gap a tie-out exists to close.

4 • The independent source

The Federal Reserve Board computes and publishes the G.19. Anyone can fetch the same table:

```
https://www.federalreserve.gov/releases/g19/hist/cc\_hist\_sa\_levels.html
```

The figure sits in the row labelled **Jun 2026**, in the column headed **Total**, under “Consumer credit outstanding, seasonally adjusted”, in millions of dollars. Here is that page, with the row ringed and the table's own column headings brought into the same frame:

Federal Reserve Board -- G.19 Consumer Credit, historical seasonally adjusted levels -- row Jun 2026				
COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE				
Date	Total	Revolving	Nonrevolving	
Jan 1943	6,577.83	n.a.	6,577.83	
Feb 1943	6,463.04	n.a.	6,463.04	
Dec 2025	5,099,419.16	1,324,322.34	3,775,096.82	
Jan 2026	5,103,845.83	1,326,083.72	3,777,762.10	
Feb 2026	5,111,066.35	1,327,206.23	3,783,860.12	
Mar 2026	5,133,899.60	1,337,959.33	3,795,940.27	
Apr 2026	5,153,816.46	1,349,631.51	3,804,184.95	
May 2026	5,152,735.06	1,344,328.87	3,808,406.19	
Jun 2026	5,166,907.71	1,351,069.14	3,815,838.57	
n.a.=not available. ...=not applicable.				
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Federal Reserve Board, G.19 historical table. Last updated 7 August 2026, per the line under the table.

5 · The comparison

ours	Raw_Consumer B1980, read from the workbook	5166907.71
source	G.19 historical SA levels, Jun 2026, 'Total'	5166907.71
diff		0.0

Same entity (all US consumer credit), same period (June 2026), same basis (seasonally adjusted, outstanding level), same units (millions of dollars). Verdict **TIED**, read off that block rather than written before it.

The sets, and the sources behind them

Split by who publishes the number, because that is what determines where the check has to go and how hard it is to get there.

FHFA All-Transactions house price indexes

Federal Housing Finance Agency — **67 of 67 tied**

FHFA -- hpi_at_state.csv, the agency's own quarterly file: header row plus three states at 2026 Q2 — file fhfa_state.csv, line numbers as in the file						
206	AK,	2026,	2,	494.	13	
1030	CA,	2026,	2,	974.	97	
2266	GA,	2026,	2,	662.	08	

FHFA hpi_at_state.csv, the agency's own quarterly file. Header row, then Alaska, California and Georgia at 2026 Q2.

FHFA -- hpi_at_us_and_census.csv, row USA 2026 Q2 — file fhfa_us.csv, line numbers as in the file						
2060	USA,	2026,	2,	719.	87	

FHFA hpi_at_us_and_census.csv, row USA 2026 Q2.

Every line in this set

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
AKSTHPI	FHFA All-Transactions HPI -- Alaska	Raw_Price!B732	494.13	494.13	0.0	TIED
ALSTHPI	FHFA All-Transactions HPI -- Alabama	Raw_Price!B628	548.12	548.12	0.0	TIED
ARSTHPI	FHFA All-Transactions HPI -- Arkansas	Raw_Price!B940	479.22	479.22	0.0	TIED
ATNHPIUS12420Q	FHFA HPI -- Austin-Round Rock-Georgetown, TX	Raw_Price!B7284	506.31	506.31	0.0	TIED
ATNHPIUS14454Q	FHFA HPI -- Boston, MA (div)	Raw_Price!B6764	502.13	502.13	0.0	TIED
ATNHPIUS16984Q	FHFA HPI -- Chicago-Naperville-Evanston, IL (div)	Raw_Price!B6140	296.21	296.21	0.0	TIED
ATNHPIUS19124Q	FHFA HPI -- Dallas-Plano-Irving, TX (div)	Raw_Price!B6244	426.45	426.45	0.0	TIED
ATNHPIUS19804Q	FHFA HPI -- Detroit-Dearborn-Livonia, MI (div)	Raw_Price!B7180	302.95	302.95	0.0	TIED
ATNHPIUS26420Q	FHFA HPI -- Houston-The Woodlands-Sugar Land, TX	Raw_Price!B6348	414.19	414.19	0.0	TIED
ATNHPIUS31084Q	FHFA HPI -- Los Angeles-Long Beach-Glendale, CA (div)	Raw_Price!B6036	557.38	557.38	0.0	TIED
ATNHPIUS33124Q	FHFA HPI -- Miami-Miami Beach-Kendall, FL (div)	Raw_Price!B6452	666.21	666.21	0.0	TIED
ATNHPIUS33460Q	FHFA HPI -- Minneapolis-St. Paul-Bloomington, MN-WI	Raw_Price!B7076	367.67	367.67	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
ATNHPIUS35614Q	FHFA HPI -- New York-Jersey City-White Plains, NY-NJ (div)	Raw_Price!B5932	454.16	454.16	0.0	TIED
ATNHPIUS37964Q	FHFA HPI -- Philadelphia, PA (div)	Raw_Price!B6556	421.68	421.68	0.0	TIED
ATNHPIUS38060Q	FHFA HPI -- Phoenix-Mesa-Chandler, AZ	Raw_Price!B6660	517.12	517.12	0.0	TIED
ATNHPIUS40140Q	FHFA HPI -- Riverside-San Bernardino-Ontario, CA	Raw_Price!B7388	501.05	501.05	0.0	TIED
ATNHPIUS41884Q	FHFA HPI -- San Francisco-San Mateo-Redwood City, CA (div)	Raw_Price!B6868	506.53	506.53	0.0	TIED
ATNHPIUS42644Q	FHFA HPI -- Seattle-Bellevue-Kent, WA (div)	Raw_Price!B6972	554.91	554.91	0.0	TIED
AZSTHPI	FHFA All-Transactions HPI -- Arizona	Raw_Price!B836	731.21	731.21	0.0	TIED
CASTHPI	FHFA All-Transactions HPI -- California	Raw_Price!B1044	974.97	974.97	0.0	TIED
COSTHPI	FHFA All-Transactions HPI -- Colorado	Raw_Price!B1148	853.19	853.19	0.0	TIED
CTSTHPI	FHFA All-Transactions HPI -- Connecticut	Raw_Price!B1252	736.5	736.5	0.0	TIED
DCSTHPI	FHFA All-Transactions HPI -- District of Columbia	Raw_Price!B1460	1037.72	1037.72	0.0	TIED
DESTHPI	FHFA All-Transactions HPI -- Delaware	Raw_Price!B1356	782.33	782.33	0.0	TIED
FLSTHPI	FHFA All-Transactions HPI -- Florida	Raw_Price!B1564	825.7	825.7	0.0	TIED
GASTHPI	FHFA All-Transactions HPI -- Georgia	Raw_Price!B1668	662.08	662.08	0.0	TIED
HISTHPI	FHFA All-Transactions HPI -- Hawaii	Raw_Price!B1772	982.57	982.57	0.0	TIED
IASTHPI	FHFA All-Transactions HPI -- Iowa	Raw_Price!B2188	469.23	469.23	0.0	TIED
IDSTHPI	FHFA All-Transactions HPI -- Idaho	Raw_Price!B1876	878.19	878.19	0.0	TIED
ILSTHPI	FHFA All-Transactions HPI -- Illinois	Raw_Price!B1980	565.87	565.87	0.0	TIED
INSTHPI	FHFA All-Transactions HPI -- Indiana	Raw_Price!B2084	538.71	538.71	0.0	TIED
KSSTHPI	FHFA All-Transactions HPI -- Kansas	Raw_Price!B2292	486.18	486.18	0.0	TIED
KYSTHPI	FHFA All-Transactions HPI -- Kentucky	Raw_Price!B2396	588.4	588.4	0.0	TIED
LASTHPI	FHFA All-Transactions HPI -- Louisiana	Raw_Price!B2500	381.58	381.58	0.0	TIED
MASTHPI	FHFA All-Transactions HPI -- Massachusetts	Raw_Price!B2812	1326.94	1326.94	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
MDSTHPI	FHFA All-Transactions HPI -- Maryland	Raw_Price!B2708	722.18	722.18	0.0	TIED
MESTHPI	FHFA All-Transactions HPI -- Maine	Raw_Price!B2604	1044.11	1044.11	0.0	TIED
MISTHPI	FHFA All-Transactions HPI -- Michigan	Raw_Price!B2916	584.9	584.9	0.0	TIED
MNSTHPI	FHFA All-Transactions HPI -- Minnesota	Raw_Price!B3020	613.16	613.16	0.0	TIED
MOSTHPI	FHFA All-Transactions HPI -- Missouri	Raw_Price!B3228	572.81	572.81	0.0	TIED
MSSTHPI	FHFA All-Transactions HPI -- Mississippi	Raw_Price!B3124	430.59	430.59	0.0	TIED
MTSTHPI	FHFA All-Transactions HPI -- Montana	Raw_Price!B3332	821.72	821.72	0.0	TIED
NCSTHPI	FHFA All-Transactions HPI -- North Carolina	Raw_Price!B4060	707.79	707.79	0.0	TIED
NDSTHPI	FHFA All-Transactions HPI -- North Dakota	Raw_Price!B4164	539.91	539.91	0.0	TIED
NESTHPI	FHFA All-Transactions HPI -- Nebraska	Raw_Price!B3436	551.5	551.5	0.0	TIED
NHSTHPI	FHFA All-Transactions HPI -- New Hampshire	Raw_Price!B3644	920.17	920.17	0.0	TIED
NJSTHPI	FHFA All-Transactions HPI -- New Jersey	Raw_Price!B3748	967.6	967.6	0.0	TIED
NMSTHPI	FHFA All-Transactions HPI -- New Mexico	Raw_Price!B3852	563.08	563.08	0.0	TIED
NVSTHPI	FHFA All-Transactions HPI -- Nevada	Raw_Price!B3540	647.75	647.75	0.0	TIED
NYSTHPI	FHFA All-Transactions HPI -- New York	Raw_Price!B3956	1184.26	1184.26	0.0	TIED
OHSTHPI	FHFA All-Transactions HPI -- Ohio	Raw_Price!B4268	522.21	522.21	0.0	TIED
OKSTHPI	FHFA All-Transactions HPI -- Oklahoma	Raw_Price!B4372	400.64	400.64	0.0	TIED
ORSTHPI	FHFA All-Transactions HPI -- Oregon	Raw_Price!B4476	855.4	855.4	0.0	TIED
PASTHPI	FHFA All-Transactions HPI -- Pennsylvania	Raw_Price!B4580	733.19	733.19	0.0	TIED
RISTHPI	FHFA All-Transactions HPI -- Rhode Island	Raw_Price!B4684	1064.86	1064.86	0.0	TIED
SCSTHPI	FHFA All-Transactions HPI -- South Carolina	Raw_Price!B4788	725.16	725.16	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
SDSTHPI	FHFA All-Transactions HPI -- South Dakota	Raw_Price!B4892	642.5	642.5	0.0	TIED
TNSTHPI	FHFA All-Transactions HPI -- Tennessee	Raw_Price!B4996	705.37	705.37	0.0	TIED
TXSTHPI	FHFA All-Transactions HPI -- Texas	Raw_Price!B5100	531.31	531.31	0.0	TIED
USSTHPI	FHFA All-Transactions House Price Index, United States	Raw_Price!B4	719.87	719.87	0.0	TIED
UTSTHPI	FHFA All-Transactions HPI -- Utah	Raw_Price!B5204	864.92	864.92	0.0	TIED
VASTHPI	FHFA All-Transactions HPI -- Virginia	Raw_Price!B5412	780.93	780.93	0.0	TIED
VTSTHPI	FHFA All-Transactions HPI -- Vermont	Raw_Price!B5308	902.53	902.53	0.0	TIED
WASTHPI	FHFA All-Transactions HPI -- Washington	Raw_Price!B5516	1033.33	1033.33	0.0	TIED
WISTHPI	FHFA All-Transactions HPI -- Wisconsin	Raw_Price!B5724	646.66	646.66	0.0	TIED
WVSTHPI	FHFA All-Transactions HPI -- West Virginia	Raw_Price!B5620	365.44	365.44	0.0	TIED
WYSTHPI	FHFA All-Transactions HPI -- Wyoming	Raw_Price!B5828	526.96	526.96	0.0	TIED

And the workbook cell behind each one

	A	B	C
730	AKSTHPI	FHFA All-Transactions HPI -- South Dakota	
731	date	value	
732	2026-04-01	494.13	
733	2026-01-01	494.25	
734	2025-10-01	485.22	

AKSTHPI **TIED**
source: hpi_at_state.csv row AK 2026Q2

	A	B	C
626	ALSTHPI	FHFA All-Transactions HPI -- South Dakota	
627	date	value	
628	2026-04-01	548.12	
629	2026-01-01	540.7	
630	2025-10-01	537.44	

ALSTHPI **TIED**
source: hpi_at_state.csv row AL 2026Q2

	A	B	C
938	ARSTHPI	FHFA All-Transactions HPI -- South Dakota	
939	date	value	
940	2026-04-01	479.22	
941	2026-01-01	477.36	
942	2025-10-01	475.48	

ARSTHPI **TIED**
source: hpi_at_state.csv row AR 2026Q2

	A	B	C
7282	ATNHPIUS124	FHFA HPI -- Austin, TX	
7283	date	value	
7284	2026-04-01	506.31	
7285	2026-01-01	501.68	
7286	2025-10-01	502.62	

ATNHPIUS12420Q **TIED**
source: hpi_at_metro.csv CBSA 12420 2026Q2

	A	B	C
6762	ATNHPIUS144	FHFA HPI -- Boston, MA	
6763	date	value	
6764	2026-04-01	502.13	
6765	2026-01-01	498.13	
6766	2025-10-01	494.56	

ATNHPIUS14454Q **TIED**
source: hpi_at_metro.csv CBSA 14454 2026Q2

	A	B	C
6138	ATNHPIUS169	FHFA HPI -- Chicago, IL	
6139	date	value	
6140	2026-04-01	296.21	
6141	2026-01-01	287.73	
6142	2025-10-01	284.48	

ATNHPIUS16984Q **TIED**
source: hpi_at_metro.csv CBSA 16984 2026Q2

	A	B	C
6242	ATNHPIUS19124Q FHFA HPI -- Dalla <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6243	date	value	
6244	2026-04-01	426.45	
6245	2026-01-01	427.83	
6246	2025-10-01	425.19	

ATNHPIUS19124Q **TIED**

source: hpi_at_metro.csv CBSA 19124 2026Q2

	A	B	C
7178	ATNHPIUS19804Q FHFA HPI -- Detrc <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
7179	date	value	
7180	2026-04-01	302.95	
7181	2026-01-01	296.06	
7182	2025-10-01	294.31	

ATNHPIUS19804Q **TIED**

source: hpi_at_metro.csv CBSA 19804 2026Q2

	A	B	C
6346	ATNHPIUS26420Q FHFA HPI -- Hous <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6347	date	value	
6348	2026-04-01	414.19	
6349	2026-01-01	410.91	
6350	2025-10-01	409.42	

ATNHPIUS26420Q **TIED**

source: hpi_at_metro.csv CBSA 26420 2026Q2

	A	B	C
6034	ATNHPIUS31084Q FHFA HPI -- Los A <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6035	date	value	
6036	2026-04-01	557.38	
6037	2026-01-01	558.88	
6038	2025-10-01	554.76	

ATNHPIUS31084Q **TIED**

source: hpi_at_metro.csv CBSA 31084 2026Q2

	A	B	C
6450	ATNHPIUS33124Q FHFA HPI -- Mian <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6451	date	value	
6452	2026-04-01	666.21	
6453	2026-01-01	670.57	
6454	2025-10-01	655.49	

ATNHPIUS33124Q **TIED**

source: hpi_at_metro.csv CBSA 33124 2026Q2

	A	B	C
7074	ATNHPIUS33460Q FHFA HPI -- Minn <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
7075	date	value	
7076	2026-04-01	367.67	
7077	2026-01-01	362.6	
7078	2025-10-01	359.74	

ATNHPIUS33460Q **TIED**

source: hpi_at_metro.csv CBSA 33460 2026Q2

	A	B	C
5930	ATNHPIUS35614Q FHFA HPI -- New <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
5931	date	value	
5932	2026-04-01	454.16	
5933	2026-01-01	441.94	
5934	2025-10-01	436.06	

ATNHPIUS35614Q **TIED**

source: hpi_at_metro.csv CBSA 35614 2026Q2

	A	B	C
6554	ATNHPIUS37964Q FHFA HPI -- Philat <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6555	date	value	
6556	2026-04-01	421.68	
6557	2026-01-01	414.88	
6558	2025-10-01	410.39	

ATNHPIUS37964Q **TIED**

source: hpi_at_metro.csv CBSA 37964 2026Q2

	A	B	C
6658	ATNHPIUS38060Q FHFA HPI -- Phoe <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6659	date	value	
6660	2026-04-01	517.12	
6661	2026-01-01	518.97	
6662	2025-10-01	515	

ATNHPIUS38060Q **TIED**

source: hpi_at_metro.csv CBSA 38060 2026Q2

	A	B	C
7386	ATNHPIUS40140Q FHFA HPI -- River <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
7387	date	value	
7388	2026-04-01	501.05	
7389	2026-01-01	500.03	
7390	2025-10-01	497.83	

ATNHPIUS40140Q **TIED**

source: hpi_at_metro.csv CBSA 40140 2026Q2

	A	B	C
6866	ATNHPIUS41884Q FHFA HPI -- San F <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6867	date	value	
6868	2026-04-01	506.53	
6869	2026-01-01	510.36	
6870	2025-10-01	512.99	

ATNHPIUS41884Q **TIED**

source: hpi_at_metro.csv CBSA 41884 2026Q2

	A	B	C
6970	ATNHPIUS42644Q FHFA HPI -- Seatt <i>freq=quarterly; units=index 1995Q1=100; transform=yoy...</i>		
6971	date	value	
6972	2026-04-01	554.91	
6973	2026-01-01	555.17	
6974	2025-10-01	554.02	

ATNHPIUS42644Q **TIED**

source: hpi_at_metro.csv CBSA 42644 2026Q2

	A	B	C
834	AZSTHPI FHFA All-Transact <i>freq=quarterly; units=index 1980Q1=100; transform=yoy...</i>		
835	date	value	
836	2026-04-01	731.21	
837	2026-01-01	732.7	
838	2025-10-01	728.62	

AZSTHPI **TIED**

source: hpi_at_state.csv row AZ 2026Q2

	A	B	C
1042	CASTHPI FHFA All-Transact <i>freq=quarterly; units=index 1980Q1=100; transform=yoy...</i>		
1043	date	value	
1044	2026-04-01	974.97	
1045	2026-01-01	975.22	
1046	2025-10-01	969.38	

CASTHPI **TIED**

source: hpi_at_state.csv row CA 2026Q2

	A	B	C
1146	COSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1147	date	value	
1148	2026-04-01	853.19	
1149	2026-01-01	854.32	
1150	2025-10-01	855.85	

COSTHPI **TIED**

source: hpi_at_state.csv row CO 2026Q2

	A	B	C
1250	CTSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1251	date	value	
1252	2026-04-01	736.5	
1253	2026-01-01	719.95	
1254	2025-10-01	709	

CTSTHPI **TIED**

source: hpi_at_state.csv row CT 2026Q2

	A	B	C
1458	DCSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1459	date	value	
1460	2026-04-01	1037.72	
1461	2026-01-01	1053.66	
1462	2025-10-01	1068.43	

DCSTHPI **TIED**

source: hpi_at_state.csv row DC 2026Q2

	A	B	C
1354	DESTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1355	date	value	
1356	2026-04-01	782.33	
1357	2026-01-01	774.16	
1358	2025-10-01	776.84	

DESTHPI **TIED**

source: hpi_at_state.csv row DE 2026Q2

	A	B	C
1562	FLSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1563	date	value	
1564	2026-04-01	825.7	
1565	2026-01-01	825.97	
1566	2025-10-01	820.6	

FLSTHPI **TIED**

source: hpi_at_state.csv row FL 2026Q2

	A	B	C
1666	GASTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1667	date	value	
1668	2026-04-01	662.08	
1669	2026-01-01	660.87	
1670	2025-10-01	656.96	

GASTHPI **TIED**

source: hpi_at_state.csv row GA 2026Q2

	A	B	C
1770	HISTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1771	date	value	
1772	2026-04-01	982.57	
1773	2026-01-01	944.79	
1774	2025-10-01	937.51	

HISTHPI **TIED**

source: hpi_at_state.csv row HI 2026Q2

	A	B	C
2186	IASTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
2187	date	value	
2188	2026-04-01	469.23	
2189	2026-01-01	462.07	
2190	2025-10-01	459.6	

IASTHPI **TIED**

source: hpi_at_state.csv row IA 2026Q2

	A	B	C
1874	IDSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1875	date	value	
1876	2026-04-01	878.19	
1877	2026-01-01	866.29	
1878	2025-10-01	852.7	

IDSTHPI **TIED**

source: hpi_at_state.csv row ID 2026Q2

	A	B	C
1978	ILSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
1979	date	value	
1980	2026-04-01	565.87	
1981	2026-01-01	550.76	
1982	2025-10-01	544.64	

ILSTHPI **TIED**

source: hpi_at_state.csv row IL 2026Q2

	A	B	C
2082	INSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
2083	date	value	
2084	2026-04-01	538.71	
2085	2026-01-01	529.17	
2086	2025-10-01	524.62	

INSTHPI **TIED**

source: hpi_at_state.csv row IN 2026Q2

	A	B	C
2290	KSSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
2291	date	value	
2292	2026-04-01	486.18	
2293	2026-01-01	480.62	
2294	2025-10-01	474.38	

KSSTHPI **TIED**

source: hpi_at_state.csv row KS 2026Q2

	A	B	C
2394	KYSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
2395	date	value	
2396	2026-04-01	588.4	
2397	2026-01-01	579.24	
2398	2025-10-01	573.53	

KYSTHPI **TIED**

source: hpi_at_state.csv row KY 2026Q2

	A	B	C
2498	LASTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
2499	date	value	
2500	2026-04-01	381.58	
2501	2026-01-01	377.29	
2502	2025-10-01	378.39	

LASTHPI **TIED**

source: hpi_at_state.csv row LA 2026Q2

	A	B	C
2810	MASTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
2811	date	value	
2812	2026-04-01	1326.94	
2813	2026-01-01	1311.85	
2814	2025-10-01	1302.38	

MASTHPI TIED

source: hpi_at_state.csv row MA 2026Q2

	A	B	C
2706	MDSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
2707	date	value	
2708	2026-04-01	722.18	
2709	2026-01-01	720.17	
2710	2025-10-01	714.02	

MDSTHPI TIED

source: hpi_at_state.csv row MD 2026Q2

	A	B	C
2602	MESTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
2603	date	value	
2604	2026-04-01	1044.11	
2605	2026-01-01	1034.24	
2606	2025-10-01	1027.83	

MESTHPI TIED

source: hpi_at_state.csv row ME 2026Q2

	A	B	C
2914	MISTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
2915	date	value	
2916	2026-04-01	584.9	
2917	2026-01-01	573.51	
2918	2025-10-01	569.28	

MISTHPI TIED

source: hpi_at_state.csv row MI 2026Q2

	A	B	C
3018	MNSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3019	date	value	
3020	2026-04-01	613.16	
3021	2026-01-01	603.44	
3022	2025-10-01	598.93	

MNSTHPI TIED

source: hpi_at_state.csv row MN 2026Q2

	A	B	C
3226	MOSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3227	date	value	
3228	2026-04-01	572.81	
3229	2026-01-01	566.49	
3230	2025-10-01	558.88	

MOSTHPI TIED

source: hpi_at_state.csv row MO 2026Q2

	A	B	C
3122	MSSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3123	date	value	
3124	2026-04-01	430.59	
3125	2026-01-01	424.71	
3126	2025-10-01	423.57	

MSSTHPI TIED

source: hpi_at_state.csv row MS 2026Q2

	A	B	C
3330	MTSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3331	date	value	
3332	2026-04-01	821.72	
3333	2026-01-01	818.01	
3334	2025-10-01	806	

MTSTHPI TIED

source: hpi_at_state.csv row MT 2026Q2

	A	B	C
4058	NCSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
4059	date	value	
4060	2026-04-01	707.79	
4061	2026-01-01	702.38	
4062	2025-10-01	699.67	

NCSTHPI TIED

source: hpi_at_state.csv row NC 2026Q2

	A	B	C
4162	NDSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
4163	date	value	
4164	2026-04-01	539.91	
4165	2026-01-01	523.49	
4166	2025-10-01	522.08	

NDSTHPI TIED

source: hpi_at_state.csv row ND 2026Q2

	A	B	C
3434	NESTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3435	date	value	
3436	2026-04-01	551.5	
3437	2026-01-01	545.41	
3438	2025-10-01	541.3	

NESTHPI TIED

source: hpi_at_state.csv row NE 2026Q2

	A	B	C
3642	NHSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3643	date	value	
3644	2026-04-01	920.17	
3645	2026-01-01	904.31	
3646	2025-10-01	896.32	

NHSTHPI TIED

source: hpi_at_state.csv row NH 2026Q2

	A	B	C
3746	NJSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3747	date	value	
3748	2026-04-01	967.6	
3749	2026-01-01	946.64	
3750	2025-10-01	938.09	

NJSTHPI TIED

source: hpi_at_state.csv row NJ 2026Q2

	A	B	C
3850	NMSTHPI	FHFA All-Transact freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
3851	date	value	
3852	2026-04-01	563.08	
3853	2026-01-01	556.16	
3854	2025-10-01	553.26	

NMSTHPI TIED

source: hpi_at_state.csv row NM 2026Q2

	A	B	C
3538	NVSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
3539	date	value	
3540	2026-04-01	647.75	
3541	2026-01-01	640.33	
3542	2025-10-01	635.63	

NVSTHPI **TIED**

source: hpi_at_state.csv row NV 2026Q2

	A	B	C
3954	NYSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
3955	date	value	
3956	2026-04-01	1184.26	
3957	2026-01-01	1158.31	
3958	2025-10-01	1148.31	

NYSTHPI **TIED**

source: hpi_at_state.csv row NY 2026Q2

	A	B	C
4266	OHSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4267	date	value	
4268	2026-04-01	522.21	
4269	2026-01-01	513.37	
4270	2025-10-01	508.16	

OHSTHPI **TIED**

source: hpi_at_state.csv row OH 2026Q2

	A	B	C
4370	OKSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4371	date	value	
4372	2026-04-01	400.64	
4373	2026-01-01	395.35	
4374	2025-10-01	395.54	

OKSTHPI **TIED**

source: hpi_at_state.csv row OK 2026Q2

	A	B	C
4474	ORSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4475	date	value	
4476	2026-04-01	855.4	
4477	2026-01-01	847.78	
4478	2025-10-01	842.58	

ORSTHPI **TIED**

source: hpi_at_state.csv row OR 2026Q2

	A	B	C
4578	PASTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4579	date	value	
4580	2026-04-01	733.19	
4581	2026-01-01	721.12	
4582	2025-10-01	711.95	

PASTHPI **TIED**

source: hpi_at_state.csv row PA 2026Q2

	A	B	C
4682	RISTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4683	date	value	
4684	2026-04-01	1064.86	
4685	2026-01-01	1040.33	
4686	2025-10-01	1030.74	

RISTHPI **TIED**

source: hpi_at_state.csv row RI 2026Q2

	A	B	C
4786	SCSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4787	date	value	
4788	2026-04-01	725.16	
4789	2026-01-01	716.54	
4790	2025-10-01	709.79	

SCSTHPI **TIED**

source: hpi_at_state.csv row SC 2026Q2

	A	B	C
4890	SDSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4891	date	value	
4892	2026-04-01	642.5	
4893	2026-01-01	640.46	
4894	2025-10-01	635.96	

SDSTHPI **TIED**

source: hpi_at_state.csv row SD 2026Q2

	A	B	C
4994	TNSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
4995	date	value	
4996	2026-04-01	705.37	
4997	2026-01-01	701.69	
4998	2025-10-01	697.29	

TNSTHPI **TIED**

source: hpi_at_state.csv row TN 2026Q2

	A	B	C
5098	TXSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
5099	date	value	
5100	2026-04-01	531.31	
5101	2026-01-01	528.51	
5102	2025-10-01	526.13	

TXSTHPI **TIED**

source: hpi_at_state.csv row TX 2026Q2

	A	B	C
2	USSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=index
3	date	value	
4	2026-04-01	719.87	
5	2026-01-01	712.2	
6	2025-10-01	707.22	

USSTHPI **TIED**

source: hpi_at_us_and_census.csv USA 2026Q2

	A	B	C
5202	UTSTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
5203	date	value	
5204	2026-04-01	864.92	
5205	2026-01-01	857.97	
5206	2025-10-01	852.11	

UTSTHPI **TIED**

source: hpi_at_state.csv row UT 2026Q2

	A	B	C
5410	VASTHPI	FHFA All-Transact	freq=quarterly; units=index 1980Q1=100; transform=yoy_...
5411	date	value	
5412	2026-04-01	780.93	
5413	2026-01-01	767.64	
5414	2025-10-01	762.67	

VASTHPI **TIED**

source: hpi_at_state.csv row VA 2026Q2

	A	B	C
5306	VTSTHPI	FHFA All-Transact{freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
5307	date	value	
5308	2026-04-01	902.53	
5309	2026-01-01	885.03	
5310	2025-10-01	886.28	

VTSTHPI **TIED**

source: hpi_at_state.csv row VT 2026Q2

	A	B	C
5514	WASTHPI	FHFA All-Transact{freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
5515	date	value	
5516	2026-04-01	1033.33	
5517	2026-01-01	1029.21	
5518	2025-10-01	1026.38	

WASTHPI **TIED**

source: hpi_at_state.csv row WA 2026Q2

	A	B	C
5722	WISTHPI	FHFA All-Transact{freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
5723	date	value	
5724	2026-04-01	646.66	
5725	2026-01-01	627.82	
5726	2025-10-01	621.14	

WISTHPI **TIED**

source: hpi_at_state.csv row WI 2026Q2

	A	B	C
5618	WVSTHPI	FHFA All-Transact{freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
5619	date	value	
5620	2026-04-01	365.44	
5621	2026-01-01	355.05	
5622	2025-10-01	352.84	

WVSTHPI **TIED**

source: hpi_at_state.csv row WV 2026Q2

	A	B	C
5826	WYSTHPI	FHFA All-Transact{freq=quarterly; units=index 1980Q1=100; transform=yoy_...	
5827	date	value	
5828	2026-04-01	526.96	
5829	2026-01-01	519.48	
5830	2025-10-01	510.7	

WYSTHPI **TIED**

source: hpi_at_state.csv row WY 2026Q2

Charge-off and delinquency rates at commercial banks

Federal Reserve Board — **29 of 29 tied**

Federal Reserve Board -- Delinquency rates, all commercial banks (delallsa) -- row 2026:2

COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE

Real estate loans Consumer loans
 Booked in domestic offices Leases C&I loans Agricultural loans Total loans and leases
 All Residential¹ Commercial² Farmland All Credit cards Other

Delinquency rates

All Banks, SA

	Real estate loans			Consumer loans			Leases	C&I loans	Agricultural loans	Total
	Booked in domestic offices			All Credit cards			Other	loans	loans	and leases
	All	Residential ¹	Commercial ²	Farmland	All	Credit cards	Other			
2026:2	1.68	1.86	1.53	1.33	2.62	2.85	2.33	1.19	1.27	1.10 1.42
2026:1	1.70	1.88	1.56	1.34	2.64	2.91	2.28	1.16	1.33	1.12 1.43
2025:4	1.68	1.79	1.58	1.31	2.63	2.95	2.27	1.22	1.34	1.16 1.46
2025:3	1.66	1.78	1.57	1.32	2.72	2.99	2.33	1.14	1.33	1.11 1.46
2025:2	1.67	1.78	1.57	1.33	2.75	3.04	2.38	1.10	1.29	1.16 1.49
2025:1	1.67	1.77	1.57	1.34	2.77	3.05	2.39	1.13	1.28	1.19 1.51
2024:4	1.64	1.77	1.56	1.20	2.76	3.08	2.44	1.11	1.27	1.17 1.50
2024:3	1.61	1.74	1.51	1.11	2.73	3.20	2.19	1.09	1.18	1.28 1.50
2024:2	1.56	1.73	1.42	1.07	2.72	3.22	2.19	1.18	1.15	0.94 1.48
2024:1	1.45	1.71	1.21	1.01	2.67	3.17	2.14	1.08	1.11	1.03 1.42
2023:4	1.41	1.70	1.16	0.99	2.62	3.10	2.14	1.03	1.02	0.80 1.37
2023:3	1.37	1.72	1.06	0.99	2.52	2.95	2.09	1.03	0.98	0.78 1.33
2023:2	1.27	1.72	0.85	0.91	2.37	2.75	2.01	0.95	1.00	0.81 1.25
2023:1	1.23	1.74	0.78	0.91	2.22	2.47	2.01	0.94	0.97	0.79 1.20
2022:4	1.21	1.79	0.69	1.02	2.07	2.26	1.90	0.97	1.02	0.94 1.19
2022:3	1.20	1.84	0.63	1.07	1.92	2.05	1.80	1.01	1.12	0.92 1.19
2022:2	1.32	1.96	0.72	1.20	1.80	1.83	1.79	1.00	1.03	0.94 1.22
2022:1	1.41	2.09	0.75	1.28	1.65	1.69	1.65	0.97	1.05	1.14 1.23
2021:4	1.49	2.29	0.78	1.55	1.55	1.57	1.48	1.04	1.12	1.25 1.25
2021:3	1.57	2.30	0.88	1.70	1.52	1.53	1.51	1.18	1.03	1.31 1.28
2021:2	1.69	2.47	0.93	1.80	1.53	1.59	1.50	1.32	1.06	1.48 1.35
2021:1	1.82	2.68	1.00	2.02	1.69	1.88	1.56	1.50	1.16	1.63 1.46
2020:4	1.94	2.75	1.15	2.19	1.94	2.10	1.78	1.44	1.24	1.76 1.57
2020:3	1.93	2.84	1.01	2.44	1.81	1.99	1.67	1.43	1.30	2.08 1.58
2020:2	1.76	2.54	0.92	2.59	2.00	2.45	1.64	1.55	1.28	2.17 1.53
2020:1	1.64	2.35	0.83	2.56	2.46	2.69	2.23	1.28	1.14	1.99 1.49
2019:4	1.52	2.34	0.67	2.36	2.35	2.61	2.09	1.26	1.09	2.05 1.41
2019:3	1.59	2.44	0.69	2.22	2.33	2.60	2.09	1.14	1.12	1.90 1.45
2019:2	1.68	2.60	0.67	2.31	2.35	2.59	2.12	1.09	1.07	1.83 1.49
2019:1	1.74	2.69	0.69	2.24	2.32	2.54	2.09	1.21	1.16	1.73 1.52
2018:4	1.80	2.83	0.71	2.28	2.34	2.54	2.15	1.03	0.94	1.73 1.51
2018:3	1.92	2.98	0.70	2.19	2.30	2.52	2.09	1.03	0.99	1.73 1.58
2018:2	2.02	3.22	0.72	2.14	2.25	2.50	2.02	1.03	1.08	1.71 1.64
2018:1	2.14	3.48	0.74	2.03	2.27	2.50	2.04	0.96	1.11	1.63 1.70
2017:4	2.25	3.57	0.75	2.02	2.25	2.48	2.02	1.02	1.18	1.53 1.77
2017:3	2.27	3.64	0.77	2.06	2.27	2.54	2.02	1.00	1.27	1.56 1.81
2017:2	2.29	3.68	0.77	1.95	2.22	2.49	1.97	0.97	1.35	1.51 1.83
2017:1	2.42	3.91	0.78	2.02	2.17	2.40	1.96	1.00	1.53	1.51 1.92
2016:4	2.59	4.14	0.85	1.93	2.16	2.37	1.97	1.06	1.58	1.45 2.01
2016:3	2.69	4.38	0.87	1.79	2.07	2.29	1.87	1.03	1.59	1.29 2.05

Federal Reserve Board, delinquency rates at all commercial banks, row 2026:2. Eleven loan categories across the row.

Federal Reserve Board -- Charge-off rates, all commercial banks (chgallsa) -- row 2026:2

COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE

Real estate loans Consumer loans
 Booked in domestic offices Leases C&I loans Agricultural loans Total loans and leases
 All Residential¹ Commercial² Farmland All Credit cards Other

Charge-off rates

All Banks, SA

	Real estate loans			Consumer loans			Leases	C&I loans	Agricultural loans	Total
	Booked in domestic offices			All Credit cards			Other	loans	loans	and leases
	All	Residential ¹	Commercial ²	Farmland	All	Credit cards	Other			
2026:2	0.07	0.00	0.14	0.02	2.66	3.82	1.21	0.41	0.58	0.15 0.53
2026:1	0.07	-0.00	0.17	0.02	2.65	3.82	1.17	0.36	0.59	0.13 0.56
2025:4	0.08	-0.00	0.14	0.01	2.80	4.09	1.21	0.37	0.56	0.20 0.58
2025:3	0.10	-0.00	0.18	0.02	2.88	4.18	1.23	0.35	0.57	0.14 0.63
2025:2	0.09	-0.00	0.18	0.03	2.87	4.19	1.20	0.29	0.58	0.17 0.61
2025:1	0.11	-0.00	0.22	0.01	3.00	4.44	1.22	0.25	0.55	0.20 0.64
2024:4	0.12	-0.00	0.26	0.01	2.97	4.57	1.18	0.32	0.52	0.22 0.64
2024:3	0.11	-0.01	0.25	0.11	3.00	4.69	1.19	0.32	0.55	0.16 0.68
2024:2	0.13	-0.01	0.26	0.01	2.92	4.56	1.19	0.26	0.48	0.27 0.66
2024:1	0.11	-0.00	0.20	0.03	2.80	4.40	1.16	0.21	0.44	0.13 0.62
2023:4	0.12	-0.00	0.26	0.03	2.63	4.18	1.14	0.19	0.42	0.10 0.60
2023:3	0.06	-0.00	0.17	-0.00	2.39	3.76	1.05	0.17	0.39	0.08 0.52
2023:2	0.08	0.01	0.15	0.01	2.07	3.23	1.04	0.13	0.31	-0.01 0.46
2023:1	0.04	-0.01	0.07	0.01	1.77	2.86	0.81	0.09	0.28	0.15 0.39
2022:4	0.00	-0.02	0.05	-0.02	1.63	2.51	0.87	0.09	0.23	-0.03 0.33
2022:3	0.00	-0.02	0.03	0.01	1.32	2.10	0.64	0.05	0.16	0.01 0.27
2022:2	0.01	-0.02	-0.01	-0.01	1.08	1.86	0.47	0.07	0.14	0.03 0.22
2022:1	0.01	-0.03	0.02	0.04	1.00	1.73	0.43	0.28	0.12	0.10 0.21
2021:4	0.00	-0.02	0.02	0.04	0.91	1.63	0.36	0.10	0.12	0.00 0.19
2021:3	0.00	-0.04	0.02	0.02	0.97	1.81	0.30	0.03	0.16	-0.00 0.20
2021:2	0.01	-0.04	0.05	0.01	1.22	2.38	0.29	0.19	0.21	0.10 0.26
2021:1	0.02	-0.04	0.10	0.00	1.53	2.81	0.54	0.30	0.29	0.08 0.33
2020:4	0.06	-0.03	0.11	0.03	1.50	2.68	0.55	0.42	0.42	0.24 0.40
2020:3	0.06	0.00	0.08	0.05	1.97	3.60	0.56	0.90	0.52	0.36 0.49
2020:2	0.05	0.00	0.07	0.09	2.25	3.78	0.92	0.54	0.59	0.20 0.55
2020:1	0.01	-0.03	0.02	0.08	2.28	3.72	0.93	0.25	0.56	0.23 0.53
2019:4	0.01	0.01	-0.00	0.08	2.30	3.78	0.90	0.24	0.36	0.25 0.49
2019:3	0.00	-0.04	0.04	0.07	2.36	3.75	0.96	0.17	0.44	0.16 0.52
2019:2	0.00	-0.02	0.02	0.05	2.25	3.63	0.92	0.20	0.31	0.26 0.47
2019:1	0.03	-0.00	0.04	0.09	2.24	3.68	0.87	0.19	0.30	0.25 0.47
2018:4	0.00	-0.00	0.03	0.07	2.25	3.67	0.88	0.18	0.28	0.16 0.45
2018:3	0.00	-0.02	0.03	0.07	2.28	3.69	0.86	0.16	0.26	0.35 0.46
2018:2	0.01	-0.00	0.01	0.03	2.23	3.61	0.89	0.19	0.29	0.24 0.46
2018:1	0.02	0.02	0.02	-0.02	2.22	3.65	0.88	0.13	0.30	0.18 0.46
2017:4	0.02	0.02	0.03	0.05	2.23	3.70	0.85	0.20	0.36	0.24 0.49
2017:3	0.03	0.04	0.04	0.04	2.21	3.59	0.89	0.17	0.34	0.18 0.47
2017:2	0.03	0.05	0.01	0.06	2.10	3.50	0.82	0.17	0.37	0.23 0.46
2017:1	0.04	0.05	0.01	0.03	2.19	3.45	1.05	0.13	0.34	0.17 0.47
2016:4	0.03	0.07	0.03	0.03	2.11	3.49	0.85	0.20	0.41	0.34 0.47
2016:3	0.05	0.10	0.00	0.02	1.88	3.06	0.78	0.23	0.45	0.15 0.45

Federal Reserve Board, charge-off rates at all commercial banks, row 2026:2.

Federal Reserve Board -- Delinquency rates, 100 largest banks (deltop100sa) -- row 2026:2

COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE

Real estate loans Consumer loans
 Booked in domestic offices All Credit cards Other Leases C&I loans Agricultural loans Total loans and leases
 All Residential¹ Commercial² Farmland

Delinquency rates

100 Largest Banks, SA

	Real estate loans			Consumer loans			Leases		C&I loans		Agricultural loans		Total
	Booked in domestic offices			All Credit cards			Other		loans		loans		and leases
	All	Residential ¹	Commercial ²	Farmland	All	Credit cards	Other	Leases	C&I loans	Agricultural loans	loans	and leases	
2026:2	1.84	1.97	1.70	1.44	2.61	2.74	2.43	1.18	1.17		1.52	1.41	
2026:1	1.89	1.99	1.78	1.48	2.65	2.80	2.37	1.14	1.25		1.72	1.44	
2025:4	1.87	1.90	1.83	1.39	2.61	2.83	2.36	1.17	1.26		1.71	1.47	
2025:3	1.87	1.90	1.86	1.55	2.71	2.87	2.43	1.06	1.25		1.46	1.48	
2025:2	1.90	1.91	1.89	1.51	2.74	2.92	2.47	1.00	1.18		1.84	1.52	
2025:1	1.92	1.88	1.94	1.70	2.76	2.93	2.48	1.01	1.17		1.70	1.55	
2024:4	1.91	1.91	1.97	1.51	2.76	2.97	2.54	1.02	1.19		2.26	1.56	
2024:3	1.87	1.87	1.90	1.31	2.72	3.08	2.24	0.96	1.09		2.84	1.57	
2024:2	1.83	1.87	1.78	1.34	2.70	3.10	2.25	1.08	1.06		1.70	1.55	
2024:1	1.71	1.86	1.51	0.77	2.65	3.03	2.18	0.95	1.03		2.02	1.50	
2023:4	1.67	1.85	1.44	0.75	2.59	2.96	2.19	0.94	0.94		1.19	1.44	
2023:3	1.63	1.89	1.33	0.74	2.50	2.81	2.14	0.94	0.90		0.99	1.40	
2023:2	1.51	1.90	1.05	0.77	2.34	2.61	2.08	0.89	0.94		1.15	1.32	
2023:1	1.50	1.92	0.98	0.94	2.20	2.32	2.08	0.90	0.90		1.32	1.29	
2022:4	1.46	1.97	0.83	1.30	2.04	2.10	1.97	0.91	0.91		1.72	1.26	
2022:3	1.44	2.02	0.74	1.52	1.89	1.89	1.88	1.01	1.04		1.64	1.25	
2022:2	1.58	2.15	0.86	1.92	1.78	1.70	1.85	1.00	0.96		1.40	1.28	
2022:1	1.70	2.30	0.90	2.14	1.63	1.58	1.69	0.98	0.98		1.84	1.30	
2021:4	1.79	2.53	0.89	2.41	1.52	1.47	1.51	1.04	1.07		1.76	1.33	
2021:3	1.86	2.54	0.99	2.70	1.51	1.44	1.58	1.21	1.01		1.67	1.35	
2021:2	1.98	2.74	1.02	2.66	1.52	1.51	1.55	1.37	1.07		2.13	1.43	
2021:1	2.13	2.94	1.09	3.25	1.69	1.80	1.60	1.55	1.23		2.30	1.56	
2020:4	2.26	3.03	1.27	3.01	1.91	2.02	1.82	1.49	1.28		2.49	1.67	
2020:3	2.24	3.13	1.04	3.03	1.80	1.88	1.72	1.43	1.35		2.37	1.67	
2020:2	1.95	2.75	0.88	3.05	2.01	2.36	1.70	1.57	1.31		2.91	1.59	
2020:1	1.76	2.50	0.70	2.91	2.45	2.55	2.29	1.30	1.05		2.17	1.49	
2019:4	1.67	2.51	0.56	3.07	2.30	2.48	2.15	1.30	1.03		2.27	1.44	
2019:3	1.78	2.66	0.59	2.99	2.35	2.54	2.17	1.16	1.06		2.99	1.50	
2019:2	1.90	2.83	0.56	3.03	2.35	2.48	2.20	1.09	0.98		2.51	1.52	
2019:1	2.01	2.98	0.59	3.03	2.35	2.48	2.16	1.20	1.10		2.51	1.59	
2018:4	2.09	3.14	0.60	2.95	2.34	2.50	2.21	1.02	0.84		2.57	1.57	
2018:3	2.24	3.32	0.59	3.00	2.31	2.45	2.17	1.03	0.87		2.85	1.65	
2018:2	2.39	3.61	0.61	3.09	2.28	2.45	2.08	1.01	0.97		2.96	1.72	
2018:1	2.55	3.92	0.64	2.97	2.28	2.44	2.09	0.92	1.02		2.72	1.80	
2017:4	2.69	4.00	0.64	3.15	2.24	2.43	2.06	0.99	1.11		2.52	1.89	
2017:3	2.70	4.09	0.65	3.08	2.28	2.49	2.08	0.97	1.18		2.68	1.92	
2017:2	2.73	4.13	0.64	3.03	2.24	2.46	2.01	0.95	1.28		2.53	1.95	
2017:1	2.89	4.40	0.65	3.28	2.17	2.36	1.98	1.00	1.47		2.60	2.04	
2016:4	3.10	4.67	0.70	2.84	2.15	2.34	1.97	1.05	1.49		2.44	2.14	
2016:3	3.22	4.93	0.69	2.59	2.08	2.27	1.89	1.02	1.54		2.05	2.18	

The same measure for the hundred largest banks.

Federal Reserve Board -- Delinquency rates, other banks (delothersa) -- row 2026:2

COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE

Real estate loans Consumer loans
 Booked in domestic offices All Credit cards Other Leases C&I loans Agricultural loans Total loans and leases
 All Residential¹ Commercial² Farmland

Delinquency rates

Other Banks, SA

	Real estate loans			Consumer loans			Leases		C&I loans		Agricultural loans		Total
	Booked in domestic offices			All Credit cards			Other		loans		loans		and leases
	All	Residential ¹	Commercial ²	Farmland	All	Credit cards	Other	Leases	C&I loans	Agricultural loans	loans	and leases	
2026:2	21.34	1.47	1.27	1.32	2.78	6.49	1.75	1.31	1.85		0.98	1.47	
2026:1	32	1.45	1.25	1.31	2.80	6.44	1.75	1.40	1.79		0.96	1.46	
2025:4	27	1.41	1.21	1.30	2.79	6.61	1.76	1.63	1.79		0.98	1.41	
2025:3	22	1.35	1.14	1.28	2.81	6.75	1.77	1.84	1.75		0.98	1.37	
2025:2	21.19	1.33	1.11	1.31	2.89	7.04	1.81	1.93	1.84		0.96	1.38	
2025:1	17	1.33	1.08	1.28	2.86	7.18	1.82	2.15	1.84		1.02	1.38	
2024:4	10.09	1.28	1.00	1.15	2.85	7.14	1.83	2.01	1.72		0.77	1.27	
2024:3	10.7	1.26	0.98	1.08	2.90	7.46	1.81	2.16	1.68		0.76	1.26	
2024:2	10.1	1.21	0.91	1.03	2.92	7.76	1.82	2.03	1.62		0.69	1.21	
2024:1	9.1	1.14	0.79	1.04	2.97	7.79	1.82	2.14	1.56		0.71	1.13	
2023:4	8.8	1.13	0.74	1.03	2.98	7.86	1.77	1.86	1.44		0.65	1.08	
2023:3	8.1	1.09	0.66	1.03	2.85	7.55	1.75	1.68	1.39		0.70	1.02	
2023:2	7.2	1.03	0.56	0.94	2.58	7.43	1.59	1.48	1.38		0.68	0.94	
2023:1	6.7	1.00	0.50	0.90	2.48	7.24	1.46	1.27	1.42		0.60	0.90	
2022:4	6.6	1.03	0.47	0.98	2.41	6.87	1.37	1.45	1.62		0.63	0.92	
2022:3	6.7	1.06	0.47	0.99	2.33	6.87	1.31	0.96	1.65		0.65	0.93	
2022:2	7.3	1.13	0.52	1.06	2.18	5.99	1.31	0.93	1.50		0.76	0.93	
2022:1	7.7	1.17	0.56	1.13	2.02	5.17	1.22	0.88	1.51		0.90	0.97	
2021:4	8.2	1.18	0.61	1.40	1.76	4.61	1.14	1.12	1.35		1.09	0.96	
2021:3	9.2	1.30	0.70	1.52	1.63	4.32	1.07	0.93	1.15		1.16	1.00	
2021:2	10.2	1.37	0.80	1.64	1.68	4.19	1.10	0.88	1.01		1.28	1.04	
2021:1	11.2	1.49	0.88	1.81	1.82	4.62	1.15	0.97	0.93		1.44	1.12	
2020:4	12.0	1.59	0.95	2.05	2.06	5.04	1.36	1.10	1.05		1.59	1.21	
2020:3	12.2	1.54	0.95	2.34	2.00	5.11	1.31	1.31	1.11		1.96	1.23	
2020:2	12.8	1.62	0.98	2.49	2.01	5.62	1.19	1.29	1.14		1.97	1.28	
2020:1	13.5	1.71	1.04	2.49	2.79	7.16	1.65	1.01	1.88		1.92	1.51	
2019:4	14.1	1.56	0.83	2.24	2.68	6.68	1.50	0.86	1.44		2.05	1.29	
2019:3	14.1	1.54	0.84	2.07	2.03	6.01	1.49	0.94	1.52		1.58	1.24	
2019:2	16.1	1.58	0.84	2.16	2.52	6.57	1.53	1.10	1.65		1.64	1.33	
2019:1	11.3	1.48	0.84	2.08	2.03	5.54	1.53	1.19	1.59		1.47	1.27	
2018:4	14.1	1.52	0.85	2.13	2.05	5.61	1.55	1.22	1.55		1.46	1.26	
2018:3	16.1	1.57	0.86	2.02	2.08	5.92	1.54	1.00	1.67		1.36	1.28	
2018:2	15.1	1.55	0.87	1.92	2.08	6.18	1.52	1.22	1.69		1.31	1.29	
2018:1	11.1	1.57	0.91	1.81	2.10	6.07	1.54	1.39	1.71		1.27	1.32	
2017:4	18.1	1.65	0.89	1.70	2.07	5.57	1.59	1.34	1.54		1.16	1.28	
2017:3	31.23	1.66	0.94	1.79	2.08	5.43	1.64	1.29	1.76		1.15	1.34	
2017:2	21.23	1.66	0.96	1.67	2.07	4.03	1.69	1.16	1.83		1.12	1.36	
2017:1	11.28	1.76	0.99	1.69	2.17	3.53	1.77	1.10	2.01		1.13	1.45	
2016:4	41.35	1.80	1.06	1.65	2.17	3.29	1.83	1.08	2.03		1.06	1.48	
2016:3	31.41	1.88	1.13	1.57	2.04	3.06	1.74	1.18	1.94		1.00	1.50	

And for every other bank.

Every line in this set

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
CORALACBS	Charge-Off Rate on All Loans & Leases, All Banks	Raw_Commercial!B836	0.55	0.55	0.0	TIED
CORBLACBS	Charge-Off Rate on Commercial & Industrial Loans, All Banks	Raw_Commercial!B4	0.58	0.58	0.0	TIED
CORCACBS	Charge-Off Rate on Consumer Loans, All Commercial Banks	Raw_Consumer!B628	2.66	2.66	0.0	TIED
CORCCACBS	Charge-Off Rate on Credit Card Loans, All Commercial Banks	Raw_Consumer!B4	3.82	3.82	0.0	TIED
CORCCOBS	Charge-Off Rate on Credit Card Loans, Banks Not Among 100 Larges	Raw_Consumer!B316	7.8	7.8	0.0	TIED
CORCCT100S	Charge-Off Rate on Credit Card Loans, Top 100 Banks	Raw_Consumer!B212	3.7	3.7	0.0	TIED
CORCOBS	Charge-Off Rate on Consumer Loans, Other Banks	Raw_Consumer!B1044	2.8	2.8	0.0	TIED
CORCREXFACBS	Charge-Off Rate on CRE Loans (excl. Farmland), All Banks	Raw_Commercial!B420	0.14	0.14	0.0	TIED
CORSFRMACBS	Charge-Off Rate on Single-Family Residential Mortgages, All Bank	Raw_Consumer!B1460	0	0.0	0.0	TIED
DRALACBS	Delinquency Rate on All Loans & Leases, All Banks	Raw_Commercial!B940	1.42	1.42	0.0	TIED
DRBLACBS	Delinquency Rate on Commercial & Industrial Loans, All Banks	Raw_Commercial!B108	1.27	1.27	0.0	TIED
DRBLOBS	Delinquency Rate on C&I Loans, Other Banks	Raw_Commercial!B316	1.85	1.85	0.0	TIED
DRBLT100S	Delinquency Rate on C&I Loans, Top 100 Banks	Raw_Commercial!B212	1.17	1.17	0.0	TIED
DRCCCLACBS	Delinquency Rate on Credit Card Loans, All Commercial Banks	Raw_Consumer!B108	2.85	2.85	0.0	TIED
DRCCLOBS	Delinquency Rate on Credit Card Loans, Other Banks	Raw_Consumer!B524	6.49	6.49	0.0	TIED
DRCCLT100S	Delinquency Rate on Credit Card Loans, Top 100 Banks	Raw_Consumer!B420	2.74	2.74	0.0	TIED
DRCLACBS	Delinquency Rate on Consumer Loans, All Commercial Banks	Raw_Consumer!B732	2.62	2.62	0.0	TIED
DRCLOBS	Delinquency Rate on Consumer Loans, Other Banks	Raw_Consumer!B940	2.78	2.78	0.0	TIED
DRCLT100S	Delinquency Rate on Consumer Loans, Top 100 Banks	Raw_Consumer!B836	2.61	2.61	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
DRCRELEXFACBS	Delinquency Rate on CRE Loans (excl. Farmland), All Banks	Raw_Commercial!B524	1.53	1.53	0.0	TIED
DRCRELEXFOBS	Delinquency Rate on CRE Loans (excl. Farmland), Other Banks	Raw_Commercial!B732	1.27	1.27	0.0	TIED
DRCRELEXFT100S	Delinquency Rate on CRE Loans (excl. Farmland), Top 100 Banks	Raw_Commercial!B628	1.7	1.7	0.0	TIED
DRLFRACBS	Delinquency Rate on Leases, All Commercial Banks	Raw_Consumer!B1876	1.19	1.19	0.0	TIED
DROCLACBS	Delinquency Rate on Other Consumer Loans, All Commercial Banks	Raw_Consumer!B1356	2.33	2.33	0.0	TIED
DROCLOBS	Delinquency Rate on Other Consumer Loans, Other Banks	Raw_Consumer!B1252	1.75	1.75	0.0	TIED
DROCLT100S	Delinquency Rate on Other Consumer Loans, Top 100 Banks	Raw_Consumer!B1148	2.43	2.43	0.0	TIED
DRSFRMACBS	Delinquency Rate on Single-Family Residential Mortgages, All Ban	Raw_Consumer!B1564	1.86	1.86	0.0	TIED
DRSFRMOBS	Delinquency Rate on Single-Family Residential Mortgages, Other B	Raw_Consumer!B1772	1.47	1.47	0.0	TIED
DRSFRMT100S	Delinquency Rate on Single-Family Residential Mortgages, Top 100	Raw_Consumer!B1668	1.97	1.97	0.0	TIED

And the workbook cell behind each one

	A	B	C
834	CORALACBS	Charge-Off Rate	cfreq=quarterly; units=percent; transform=zscore_8q; vinta
835	date	value	
836	2026-04-01	0.55	
837	2026-01-01	0.56	
838	2025-10-01	0.58	

CORALACBS **TIED**
source: chgallsa row 2026:2 column 11

	A	B	C
2	CORBLACBS	Charge-Off Rate	cfreq=quarterly; units=percent; transform=zscore_8q; vinta
3	date	value	
4	2026-04-01	0.58	
5	2026-01-01	0.59	
6	2025-10-01	0.56	

CORBLACBS **TIED**
source: chgallsa row 2026:2 column 9

	A	B	C
626	CORCACBS	Charge-Off Rate	cfreq=quarterly; units=percent; transform=zscore_8q; vinta
627	date	value	
628	2026-04-01	2.66	
629	2026-01-01	2.65	
630	2025-10-01	2.8	

CORCACBS **TIED**
source: chgallsa row 2026:2 column 5

	A	B	C
2	CORCCACBS	Charge-Off Rate	cfreq=quarterly; units=percent; transform=zscore_8q; vinta
3	date	value	
4	2026-04-01	3.82	
5	2026-01-01	3.82	
6	2025-10-01	4.09	

CORCCACBS **TIED**
source: chgallsa row 2026:2 column 6

	A	B	C
314	CORCCOBS	Charge-Off Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
315	date	value	
316	2026-04-01	7.8	
317	2026-01-01	8.02	
318	2025-10-01	8.25	

CORCCOBS **TIED**

source: chgothersa row 2026:2 column 6

	A	B	C
210	CORCCT100S	Charge-Off Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
211	date	value	
212	2026-04-01	3.7	
213	2026-01-01	3.66	
214	2025-10-01	4.02	

CORCCT100S **TIED**

source: chgtop100sa row 2026:2 column 6

	A	B	C
1042	CORCOBS	Charge-Off Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
1043	date	value	
1044	2026-04-01	2.8	
1045	2026-01-01	2.73	
1046	2025-10-01	3.15	

CORCOBS **TIED**

source: chgothersa row 2026:2 column 5

	A	B	C
418	CORCREXFAC	Charge-Off Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
419	date	value	
420	2026-04-01	0.14	
421	2026-01-01	0.17	
422	2025-10-01	0.14	

CORCREXFACBS **TIED**

source: chgallsa row 2026:2 column 3

	A	B	C
1458	CORSFRMACBS	Charge-Off Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
1459	date	value	
1460	2026-04-01	0	
1461	2026-01-01	0	
1462	2025-10-01	0	

CORSFRMACBS **TIED**

source: chgallsa row 2026:2 column 2

	A	B	C
938	DRALACBS	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
939	date	value	
940	2026-04-01	1.42	
941	2026-01-01	1.45	
942	2025-10-01	1.46	

DRALACBS **TIED**

source: delallsa row 2026:2 column 11

	A	B	C
106	DRBLACBS	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
107	date	value	
108	2026-04-01	1.27	
109	2026-01-01	1.33	
110	2025-10-01	1.34	

DRBLACBS **TIED**

source: delallsa row 2026:2 column 9

	A	B	C
314	DRBLOBS	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
315	date	value	
316	2026-04-01	1.85	
317	2026-01-01	1.79	
318	2025-10-01	1.79	

DRBLOBS **TIED**

source: delothersa row 2026:2 column 9

	A	B	C
210	DRBLT100S	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
211	date	value	
212	2026-04-01	1.17	
213	2026-01-01	1.25	
214	2025-10-01	1.26	

DRBLT100S **TIED**

source: deltop100sa row 2026:2 column 9

	A	B	C
106	DRCCLACBS	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
107	date	value	
108	2026-04-01	2.85	
109	2026-01-01	2.91	
110	2025-10-01	2.95	

DRCCLACBS **TIED**

source: delallsa row 2026:2 column 6

	A	B	C
522	DRCCLOBS	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
523	date	value	
524	2026-04-01	6.49	
525	2026-01-01	6.44	
526	2025-10-01	6.61	

DRCCLOBS **TIED**

source: delothersa row 2026:2 column 6

	A	B	C
418	DRCCLT100S	Delinquency Rate freq=quarterly; units=percent; transform=zscore_8q; vinta	
419	date	value	
420	2026-04-01	2.74	
421	2026-01-01	2.8	
422	2025-10-01	2.83	

DRCCLT100S **TIED**

source: deltop100sa row 2026:2 column 6

	A	B	C
730	DRCLACBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
731	date	value	
732	2026-04-01	2.62	
733	2026-01-01	2.64	
734	2025-10-01	2.63	

DRCLACBS **TIED**

source: delallsa row 2026:2 column 5

	A	B	C
938	DRCLOBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
939	date	value	
940	2026-04-01	2.78	
941	2026-01-01	2.8	
942	2025-10-01	2.79	

DRCLOBS **TIED**

source: delothersa row 2026:2 column 5

	A	B	C
834	DRCLT100S	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
835	date	value	
836	2026-04-01	2.61	
837	2026-01-01	2.63	
838	2025-10-01	2.61	

DRCLT100S **TIED**

source: deltop100sa row 2026:2 column 5

	A	B	C
522	DRCRELEXFA	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
523	date	value	
524	2026-04-01	1.53	
525	2026-01-01	1.56	
526	2025-10-01	1.58	

DRCRELEXFACBS **TIED**

source: delallsa row 2026:2 column 3

	A	B	C
730	DRCRELEXFO	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
731	date	value	
732	2026-04-01	1.27	
733	2026-01-01	1.25	
734	2025-10-01	1.21	

DRCRELEXFOBS **TIED**

source: delothersa row 2026:2 column 3

	A	B	C
626	DRCRELEXFT	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
627	date	value	
628	2026-04-01	1.7	
629	2026-01-01	1.78	
630	2025-10-01	1.83	

DRCRELEXFT100S **TIED**

source: deltop100sa row 2026:2 column 3

	A	B	C
1874	DRLFRACBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1875	date	value	
1876	2026-04-01	1.19	
1877	2026-01-01	1.16	
1878	2025-10-01	1.22	

DRLFRACBS **TIED**

source: delallsa row 2026:2 column 8

	A	B	C
1354	DROCLACBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1355	date	value	
1356	2026-04-01	2.33	
1357	2026-01-01	2.28	
1358	2025-10-01	2.27	

DROCLACBS **TIED**

source: delallsa row 2026:2 column 7

	A	B	C
1250	DROCLOBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1251	date	value	
1252	2026-04-01	1.75	
1253	2026-01-01	1.75	
1254	2025-10-01	1.76	

DROCLOBS **TIED**

source: delothersa row 2026:2 column 7

	A	B	C
1146	DROCLT100S	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1147	date	value	
1148	2026-04-01	2.43	
1149	2026-01-01	2.37	
1150	2025-10-01	2.36	

DROCLT100S **TIED**

source: deltop100sa row 2026:2 column 7

	A	B	C
1562	DRSFRMACBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1563	date	value	
1564	2026-04-01	1.86	
1565	2026-01-01	1.88	
1566	2025-10-01	1.79	

DRSFRMACBS **TIED**

source: delallsa row 2026:2 column 2

	A	B	C
1770	DRSFRMOBS	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1771	date	value	
1772	2026-04-01	1.47	
1773	2026-01-01	1.45	
1774	2025-10-01	1.41	

DRSFRMOBS **TIED**

source: delothersa row 2026:2 column 2

	A	B	C
1666	DRSFRMT100S	Delinquency Rate	<i>freq=quarterly; units=percent; transform=zscore_8q; vinta</i>
1667	date	value	
1668	2026-04-01	1.97	
1669	2026-01-01	1.99	
1670	2025-10-01	1.9	

DRSFRMT100S **TIED**

source: deltop100sa row 2026:2 column 2

G.19 consumer credit, the debt service ratios, FHFA monthly

Federal Reserve Board and FHFA — 9 of 9 tied

Federal Reserve Board -- G.19 Consumer Credit, historical seasonally adjusted levels -- row Jun 2026				
COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE				
Date	Total	Revolving	Nonrevolving	
Jan 1943	6,577.83	n.a.	6,577.83	
Feb 1943	6,463.04	n.a.	6,463.04	
Dec 2025	5,099,419.16	1,324,322.34	3,775,096.82	
Jan 2026	5,103,845.83	1,326,083.72	3,777,762.10	
Feb 2026	5,111,066.35	1,327,206.23	3,783,860.12	
Mar 2026	5,133,899.60	1,337,959.33	3,795,940.27	
Apr 2026	5,153,816.46	1,349,631.51	3,804,184.95	
May 2026	5,152,735.06	1,344,328.87	3,808,406.19	
Jun 2026	5,166,907.71	1,351,069.14	3,815,838.57	
n.a.=not available. ...=not applicable.				
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G.19 historical table, seasonally adjusted levels, row Jun 2026. Total, revolving and nonrevolving, to the cent.

Federal Reserve Board -- G.19 Consumer Credit, current release -- row 'Total outstanding', last column is June 2026														
COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE														
	2025							2026						
	2021	2022	2023	2024	2025 [†]	Q2	Q3	Q4	Q1 [†]	Q2 [†]	Apr [†]	May [†]	Jun [†]	
Total percent change (annual rate) ²	5.7	7.7	2.7	2.0	2.2	2.82	3.2	1.2	2.6	4.7	-0.3	3.3		
Total percent change (annual rate) ²	5.7	7.7	2.7	2.0	2.2	2.8	2.5	2.1	2.7	2.6	4.7	-0.3	3.3	
Revolving	6.9	15.4	8.9	4.1	3.1	3.0	3.4	3.5	4.1	3.9	10.5	-4.7	6.0	
Nonrevolving ³	5.4	5.4	0.6	1.2	1.8	2.8	1.9	1.6	2.2	2.1	2.6	1.3	2.3	
Total flow (annual rate) ^{2,d}	237.8	345.7	129.8	98.9	107.0	141.8	113.9	107.0	137.9	132.0	239.0	-13.0	170.1	
Revolving	65.7	159.1	106.2	52.9	40.0	38.1	43.9	46.4	54.5	52.4	140.1	-63.6	80.9	
Nonrevolving ³	172.2	186.6	23.6	45.9	67.0	103.6	70.0	60.6	83.4	79.6	98.9	50.7	89.2	
Total outstanding	4,512.7	4,858.4	4,988.2	4,948.1	5,099.4	5,044.2	5,072.7	5,099.4	5,133.9	5,166.9	5,153.8	5,152.7	5,166.9	
Revolving	1,033.5	1,192.6	1,298.9	1,297.0	1,324.3	1,301.7	1,312.7	1,324.3	1,338.0	1,351.1	1,349.6	1,344.3	1,351.1	
Nonrevolving ³	3,479.2	3,665.8	3,689.4	3,651.1	3,775.1	3,742.4	3,759.9	3,775.1	3,795.9	3,815.8	3,804.2	3,808.4	3,815.8	
Terms of Credit														
Not seasonally adjusted. Percent except as noted.														
Commercial bank interest rates ⁵														
New car loans														
60-month	4.82	5.36	7.83	8.16	7.65	7.67	7.64	7.24	7.53	7.14	n.a.	7.14	n.a.	
72-month	4.82	5.50	7.89	8.29	7.80	7.73	7.80	7.50	7.53	6.97	n.a.	6.97	n.a.	
Credit card plans														
All accounts	14.60	16.26	20.90	21.58	21.22	21.16	21.39	20.97	21.00	20.94	n.a.	20.94	n.a.	
Accounts assessed interest	16.45	17.91	22.15	22.89	22.32	22.25	22.83	22.30	21.52	22.15	n.a.	22.15	n.a.	
Personal loans														
24-month	9.38	9.87	11.87	12.27	11.50	11.57	11.14	11.63	11.36	11.86	n.a.	11.86	n.a.	
Finance companies (new car loans) ⁶														
Interest rates	4.6	5.2	6.7	6.2	6.4	6.6	6.3	6.1	6.1	n.a.	n.a.	n.a.	n.a.	
Maturity (months)	67	67	66	66	66	66	66	66	66	n.a.	n.a.	n.a.	n.a.	
Amount financed (dollars)	35,307	38,900	38,716	39,386	40,582	39,867	40,803	41,742	42,504	n.a.	n.a.	n.a.	n.a.	
Consumer Credit Outstanding (Levels) ¹														
Not seasonally adjusted. Billions of dollars.														
Total	2025							2026						
	2021	2022	2023	2024	2025	Q2	Q3	Q4	Q1	Q2 [†]	Apr [†]	May [†]	Jun [†]	
Total	4,512.7	4,858.4	4,988.2	4,948.1	5,099.4	4,996.3	5,040.3	5,099.4	5,074.6	5,119.0	5,096.2	5,103.4	5,119.0	
Major holders														
Depository institutions	1,827.2	2,032.8	2,117.6	2,025.2	2,074.2	1,989.5	2,013.4	2,074.2	2,034.8	2,082.9	2,056.9	2,067.3	2,082.9	
Finance companies	695.1	682.7	727.3	739.4	713.6	724.8	717.0	713.6	705.0	699.5	702.7	701.1	699.5	

The current G.19 release, which carries the percent change and the unadjusted level as well, rounded to a tenth of a billion.

Federal Reserve Board -- Household Debt Service Ratio (credit-bureau methodology) -- row 2026:1

COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE

Quarter	DSR	
	Total	Mortgage Consumer
2005:1	14.80	7.50 7.30
2023:4	11.10	5.64 5.40
2024:1	11.06	5.66 5.40
2024:2	11.02	5.70 5.32
2024:3	11.14	5.74 5.39
2024:4	11.12	5.69 5.43
2025:1	11.11	5.76 5.34
2025:2	11.12	5.83 5.30
2025:3	11.23	5.87 5.36
2025:4	11.32	5.92 5.40
2026:1	11.16	5.88 5.29

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Board of Governors of the Federal Reserve System

The debt service ratios, row 2026:1, on the credit-bureau methodology in force since the 2024:Q2 publication.

Federal Reserve Board -- the ARCHIVED debt service page, frozen at 2024:1. Reading this one would have compared a 2026 figure against a discontinued series.

2022:4	14.40	9.89	4.01	5.88
2023:1	14.12	9.71	3.94	5.77
2023:2	14.11	9.70	3.93	5.77
2023:3	14.20	9.78	3.99	5.79
2023:4	ND	9.83	4.03	5.81
2024:1	ND	9.78	4.00	5.78

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Last update: September 6, 2024

The page that would have been the wrong source: an archive frozen at 2024:1, last updated 6 September 2024, computed a different way.

Every line in this set

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
CDSP	Consumer Debt Service Payments as % of Disposable Income	Raw_Consumer!B2708	5.287795	5.29	-0.002205	TIED
HIIPONM226S	FHFA Purchase-Only House Price Index, United States (Monthly, SA	Raw_Price!B108	442.53	442.53	0.0	TIED
MDSP	Mortgage Debt Service Payments as % of Disposable Income	Raw_Consumer!B2604	5.876343	5.88	-0.003657	TIED
NONREVSL	Nonrevolving Consumer Credit Owned and Securitized (SA)	Raw_Consumer!B2292	3815838.57	3815838.57	0.0	TIED
REVOLSL	Revolving Consumer Credit Owned and Securitized (SA)	Raw_Consumer!B2188	1351069.14	1351069.14	0.0	TIED
TDSP	Household Debt Service Payments as % of Disposable Income	Raw_Consumer!B2500	11.164138	11.16	0.004138	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
TOTALNS	Total Consumer Credit Owned and Securitized (NSA)	Raw_Consumer!B2084	5119024.09	5119000.0	24.09	TIED
TOTALSL	Total Consumer Credit Owned and Securitized (SA)	Raw_Consumer!B1980	5166907.71	5166907.71	0.0	TIED
TOTALSLAR	Total Consumer Credit, Percent Change at Annual Rate	Raw_Consumer!B2396	3.3	3.3	0.0	TIED

And the workbook cell behind each one

	A	B	C
2706	CDSP	Consumer Debt S	<i>freq=quarterly; units=percent; transform=level; vintage=20</i>
2707	date	value	
2708	2026-01-01	5.287795	
2709	2025-10-01	5.400121	
2710	2025-07-01	5.355717	

CDSP **TIED**
source: row '2026:1', column 'Consumer DSR'

	A	B	C
106	HPIPONM226S	FHFA Purchase-O	<i>freq=monthly; units=index 1991-01=100; transform=index</i>
107	date	value	
108	2026-06-01	442.53	
109	2026-05-01	442.42	
110	2026-04-01	440.97	

HPIPONM226S **TIED**
source: hpi_po_monthly_hist sheet 'HPI_PO_monthly_hist', row 2026-06, column 'USA (SA)' (column 21 of 21)

	A	B	C
2602	MDSP	Mortgage Debt S	<i>freq=quarterly; units=percent; transform=level; vintage=20</i>
2603	date	value	
2604	2026-01-01	5.876343	
2605	2025-10-01	5.922638	
2606	2025-07-01	5.873735	

MDSP **TIED**
source: row '2026:1', column 'Mortgage DSR'

	A	B	C
2290	NONREVSL	Nonrevolving Cot	<i>freq=monthly; units=billions \$; transform=yoy_pct; vintage</i>
2291	date	value	
2292	2026-06-01	3815838.57	
2293	2026-05-01	3808406.19	
2294	2026-04-01	3804184.95	

NONREVSL **TIED**
source: Jun 2026 row, column 'Nonrevolving'

	A	B	C
2186	REVOLSL	Revolving Consur	<i>freq=monthly; units=billions \$; transform=yoy_pct; vintage</i>
2187	date	value	
2188	2026-06-01	1351069.14	
2189	2026-05-01	1344328.87	
2190	2026-04-01	1349631.51	

REVOLSL **TIED**
source: Jun 2026 row, column 'Revolving'

	A	B	C
2498	TDSP	Household Debt	<i>freq=quarterly; units=percent; transform=level; vintage=20</i>
2499	date	value	
2500	2026-01-01	11.164138	
2501	2025-10-01	11.322759	
2502	2025-07-01	11.229451	

TDSP **TIED**
source: row '2026:1', column 'DSR'

	A	B	C
2082	TOTALNS	Total Consumer C	<i>freq=monthly; units=billions \$; transform=yoy_pct; vintage</i>
2083	date	value	
2084	2026-06-01	5119024.09	
2085	2026-05-01	5103400.65	
2086	2026-04-01	5096188.74	

TOTALNS **TIED**
source: table 'Consumer Credit Outstanding (Levels), not seasonally adjusted, billions of dollars', row 'Total', column 'Jun 2026 p'

	A	B	C
1978	TOTALSL	Total Consumer C	<i>freq=monthly; units=billions \$; transform=yoy_pct; vintage</i>
1979	date	value	
1980	2026-06-01	5166907.71	
1981	2026-05-01	5152735.06	
1982	2026-04-01	5153816.46	

TOTALSL **TIED**
source: Jun 2026 row, column 'Total'

	A	B	C
2394	TOTALSLAR	Total Consumer C (freq=monthly; units=percent (annual rate); transform=level)	
2395	date	value	
2396	2026-06-01	3.3	
2397	2026-05-01	-0.25	
2398	2026-04-01	4.66	

TOTALSLAR **TIED**

source: table 'Consumer Credit Outstanding, seasonally adjusted', row 'Total percent change (annual rate)', column 'Jun 2026 p'

Senior Loan Officer Opinion Survey

Federal Reserve Board — 13 of 13 tied

Federal Reserve Board -- Senior Loan Officer Opinion Survey, chart data -- row 2026:3									
COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE									
Period	Panel 1: Net Percentage of Domestic Respondents Tightening Standards for C&I Loans			Panel 2: Net Percentage of Domestic Respondents Increasing Spreads of Loan Rates over Banks' Cost of Funds			Panel 3: Net Percentage of Domestic Respondents Reporting Stronger Demand for C&I Loans		
	Large and medium		Small	Large and medium		Small	Large and medium		Small
1990:2 54.4			52.7	10.7		7.4	n.a.		n.a.
2025:1 6.2			11.1	-9.4		-8.2	9.4		3.2
2025:2 18.5			15.9	-4.8		-6.6	-20.3		-20.6
2025:3 9.5			8.2	-9.5		-3.4	-28.6		-27.9
2025:4 6.5			8.3	-16.4		-6.8	11.5		-1.7
2026:1 5.3			8.9	-14.3		0.0	16.1		0.0
2026:2 8.1			6.6	-12.9		-8.2	4.8		0.0
2026:3 0.0			1.8	-26.8		-14.5	16.1		3.6

Figure 2: Measures of Supply and Demand for Commercial Real Estate Loans

Period	Panel 1: Net Percentage of Domestic Respondents Tightening Standards for Commercial Real Estate Loans ^a				Panel 2: Net Percentage of Domestic Respondents Reporting Stronger Demand for Commercial Real Estate Loans ^a			
	All	Construction and land development	Nonfarm nonresidential	Multifamily	All	Construction and land development	Nonfarm nonresidential	Multifamily
1990:2 n.a.	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1990:3 69.8	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1990:4 61.2	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1991:1 46.4	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1991:2 24.1	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1991:3 27.2	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1991:4 15.4	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1992:1 7.5	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1992:2 10.3	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1992:3 4.4	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1992:4 3.6	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1993:1 1.8	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1993:2 2.2	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1993:3 -0.9	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1993:4 -3.9	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1994:1 -4.7	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1994:2 -7.1	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1994:3 -3.1	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1994:4 -3.1	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1995:1 0.5	n.a.		n.a.	n.a.	n.a.		n.a.	n.a.
1995:2 -0.4	n.a.		n.a.	n.a.	13.8	n.a.	n.a.	n.a.
1995:3 3.1	n.a.		n.a.	n.a.	20.7	n.a.	n.a.	n.a.
1995:4 6.3	n.a.		n.a.	n.a.	16.4	n.a.	n.a.	n.a.
1996:1 12.3	n.a.		n.a.	n.a.	8.9	n.a.	n.a.	n.a.
2026:3 0.0					1.6			

The Board's SLOOS chart data, row 2026:3, with the panel headings copied in from the top of the same table.

COLUMN HEADINGS, COPIED FROM THE TOP OF THIS SAME TABLE

All Respondents Large Banks Other Banks

A. Standards for large and middle-market firms (annual sales of \$50 million or more):

B. Standards for **small firms** (annual sales of less than \$50 million).

For this question, 1 respondent answered "My bank does not originate C&I loans or credit lines to small firms."

A. Terms for large and middle-market firms (annual sales of \$50 million or more):

a. Maximum size of credit lines

b. Maximum maturity of loans or credit lines

Every line in this set

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
DRSDCILM	Net % Reporting Stronger Demand for C&I Loans, Large/Medium Firm	Raw_Commercial!B1252	16.1	16.1	0.0	TIED
DRSDCIS	Net % Reporting Stronger Demand for C&I Loans, Small Firms	Raw_Commercial!B1356	3.6	3.6	0.0	TIED
DRTSCILM	Net % Tightening Standards for C&I Loans to Large/Medium Firms	Raw_Commercial!B1044	0	0.0	0.0	TIED
DRTSCIS	Net % Tightening Standards for C&I Loans to Small Firms	Raw_Commercial!B1148	1.8	1.8	0.0	TIED
DRTSCLCC	Net % Banks Tightening Standards for Credit Card Loans	Raw_Consumer!B2812	6.7	6.7	0.0	TIED
DRTSSP	Net % Banks Tightening Standards for Consumer Loans (subprime)	Raw_Consumer!B3124	9.1	9.1	0.0	TIED
STDSAUTO	Net % Banks Tightening Standards for Auto Loans	Raw_Consumer!B2916	-4.5	-4.5	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
STDSOTHCONS	Net % Banks Tightening Standards for Other Consumer Loans	Raw_Consumer!B3020	0	0.0	0.0	TIED
SUBLPDCILSLGNQ	Net % Increasing Spreads on C&I Loans to Large/Medium Firms	Raw_Commercial1!B1772	4.5	4.5	0.0	TIED
SUBLPDHMTSENQ	Net % Banks Reporting Stronger Demand for Consumer Loans	Raw_Consumer!B3228	0	0.0	0.0	TIED
SUBLPDRCS	Net % Tightening Standards for CRE Nonfarm Nonresidential Loans	Raw_Commercial1!B1668	-3.7	-3.7	0.0	TIED
SUBLPDRCSM	Net % Tightening Standards for CRE Multifamily Loans	Raw_Commercial1!B1564	-5.7	-5.7	0.0	TIED
SUBLPDRCSN	Net % Tightening Standards for CRE Construction & Land Loans	Raw_Commercial1!B1460	-11.3	-11.3	0.0	TIED

And the workbook cell behind each one

	A	B	C
1250	DRSDCILM	Net % Reporting	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1251	date	value	
1252	2026-07-01	16.1	
1253	2026-04-01	4.8	
1254	2026-01-01	16.1	

DRSDCILM **TIED**

source: table 1, Panel 3: stronger demand for C&I loans, column 'Large and medium', row 2026:3

	A	B	C
1354	DRSDCIS	Net % Reporting	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1355	date	value	
1356	2026-07-01	3.6	
1357	2026-04-01	0	
1358	2026-01-01	0	

DRSDCIS **TIED**

source: table 1, Panel 3: stronger demand for C&I loans, column 'Small', row 2026:3

	A	B	C
1042	DRTSCILM	Net % Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1043	date	value	
1044	2026-07-01	0	
1045	2026-04-01	8.1	
1046	2026-01-01	5.3	

DRTSCILM **TIED**

source: table 1, Panel 1: tightening standards for C&I loans, column 'Large and medium', row 2026:3

	A	B	C
1146	DRTSCIS	Net % Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1147	date	value	
1148	2026-07-01	1.8	
1149	2026-04-01	6.6	
1150	2026-01-01	8.9	

DRTSCIS **TIED**

source: table 1, Panel 1: tightening standards for C&I loans, column 'Small', row 2026:3

	A	B	C
2810	DRTSCLCC	Net % Banks Tigh	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
2811	date	value	
2812	2026-07-01	6.7	
2813	2026-04-01	2	
2814	2026-01-01	0	

DRTSCLCC **TIED**

source: table 4, Panel 1: tightening standards on consumer loans, column 'Credit cards', row 2026:3

	A	B	C
3122	DRTSSP	Net % Banks Tigh	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
3123	date	value	
3124	2026-07-01	9.1	
3125	2026-04-01	6.3	
3126	2026-01-01	8.3	

DRTSSP **TIED**

source: table 3, Panel 1: tightening standards for mortgage loans, column 'Subprime', row 2026:3

	A	B	C
2914	STDSAUTO	Net % Banks Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
2915	date	value	
2916	2026-07-01	-4.5	
2917	2026-04-01	-2	
2918	2026-01-01	-6.1	

STDSAUTO **TIED**

source: table 4, Panel 1: tightening standards on consumer loans, column 'New and used autos', row 2026:3

	A	B	C
3018	STDSOTHCON	Net % Banks Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
3019	date	value	
3020	2026-07-01	0	
3021	2026-04-01	5.5	
3022	2026-01-01	2	

STDSOTHCONS **TIED**

source: table 4, Panel 1: tightening standards on consumer loans, column 'Consumer loans excluding credit cards and autos', row 2026:3

	A	B	C
1770	SUBLPDCILSL	Net % Increasing	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1771	date	value	
1772	2026-07-01	4.5	
1773	2026-04-01	10	
1774	2026-01-01	-5.3	

SUBLPDCILSLGNQ **TIED**

source: survey Table 1, question 1 (standards on C&I loans to large and middle-market firms), 'Large Banks' percent column: eased considerably 0.0 eased somewhat

	A	B	C
3226	SUBLPDH MSE	Net % Banks Repaying	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
3227	date	value	
3228	2026-07-01	0	
3229	2026-04-01	-3.7	
3230	2026-01-01	-5.7	

SUBLPDH MSENQ **TIED**

source: table 3, Panel 1: tightening standards for mortgage loans, column 'GSE-eligible', row 2026:3

	A	B	C
1666	SUBLPDR CSC	Net % Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1667	date	value	
1668	2026-07-01	-3.7	
1669	2026-04-01	4.9	
1670	2026-01-01	1.8	

SUBLPDR CSC **TIED**

source: table 2, Panel 1: tightening standards for CRE loans, column 'Construction and land development', row 2026:3

	A	B	C
1562	SUBLPDR CS M	Net % Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1563	date	value	
1564	2026-07-01	-5.7	
1565	2026-04-01	0	
1566	2026-01-01	-5.5	

SUBLPDR CS M **TIED**

source: table 2, Panel 1: tightening standards for CRE loans, column 'Multifamily', row 2026:3

	A	B	C
1458	SUBLPDR CS N	Net % Tightening	<i>freq=quarterly; units=net percent; transform=level; vintag</i>
1459	date	value	
1460	2026-07-01	-11.3	
1461	2026-04-01	-3.3	
1462	2026-01-01	-3.6	

SUBLPDR CS N **TIED**

source: table 2, Panel 1: tightening standards for CRE loans, column 'Nonfarm nonresidential', row 2026:3

S&P Cotality Case-Shiller house price indexes

S&P Dow Jones Indices — **22 of 22 tied**

S&P Dow Jones Indices refuses scripted requests outright — every attempt returned `HTTP 403`. The obstacle was real. It was not the verdict: S&P publishes a free monthly release, and that release carries the unadjusted index level for every metro and the seasonally adjusted month-on-month change for every metro. Twenty of these twenty-two series are the adjusted *levels*, which the release does not print — but two adjacent cells of our own workbook imply the change, and the change is published.

S&P Dow Jones Indices

A Division of **S&P Global**

PRESS RELEASE

Table 3 below shows a summary of the monthly changes using the seasonally adjusted (SA) and non-seasonally adjusted (NSA) data. Since its launch in early 2006, the S&P Cotality Case-Shiller Indices have published, and the markets have followed and reported on, the non-seasonally adjusted data set used in the headline indices. For analytical purposes, S&P Dow Jones Indices publishes a seasonally adjusted data set covered in the headline indices, as well as for the 17 of 20 markets with tiered price indices and the five condo markets that are tracked.

Metropolitan Area	June / May Change (%)		May / April Change (%)	
	NSA	SA	NSA	SA
Atlanta	0.24%	-0.12%	0.67%	0.00%
Boston	0.32%	0.11%	1.73%	0.55%
Charlotte	0.44%	-0.03%	0.56%	-0.24%
Chicago	0.95%	0.35%	1.60%	0.54%
Cleveland	0.98%	0.48%	1.08%	-0.13%
Dallas	0.31%	-0.07%	1.01%	0.06%
Denver	0.19%	0.10%	0.24%	-0.33%
Detroit	--	--	1.11%	0.15%
Las Vegas	-0.07%	-0.50%	0.29%	-0.55%
Los Angeles	0.38%	0.39%	0.59%	0.28%
Miami	0.32%	-0.10%	0.61%	0.09%
Minneapolis	0.70%	0.15%	1.09%	-0.10%
New York	1.03%	0.72%	1.39%	0.50%
Phoenix	-0.10%	-0.40%	0.46%	-0.22%
Portland	0.27%	0.11%	0.51%	-0.21%
San Diego	-0.14%	-0.11%	-0.17%	-0.53%
San Francisco	-0.08%	0.31%	0.84%	0.21%
Seattle	-0.47%	-0.34%	0.47%	-0.27%
Tampa	0.72%	0.26%	0.66%	-0.22%
Washington	0.40%	0.41%	0.75%	0.13%
Composite-10	0.52%	0.35%	0.95%	0.30%
Composite-20	0.43%	0.24%	0.89%	0.18%
U.S. National	0.37%	0.13%	0.70%	0.02%

Sources: S&P Dow Jones Indices and Cotality
Data through June 2006

S&P Dow Jones Indices, June 2006 release, Table 3: the seasonally adjusted month-on-month change for every metro. This is what the twenty adjusted series are proved against.

S&P Dow Jones Indices

A Division of S&P Global

PRESS RELEASE

Table 2 below summarizes the results for June 2026. The S&P Cotality Case-Shiller Indices could be revised for the prior 24 months, based on the receipt of additional source data.

Metropolitan Area	June 2026 Level	June / May Change (%)	May / April Change (%)	1-Year Change (%)
Atlanta	252.71	0.24%	0.67%	0.27%
Boston	363.57	0.32%	1.73%	2.71%
Charlotte	289.70	0.44%	0.56%	0.52%
Chicago	237.47	0.95%	1.60%	6.90%
Cleveland	210.60	0.98%	1.08%	4.13%
Dallas	297.16	0.31%	1.01%	-0.66%
Denver	317.59	0.19%	0.24%	-1.24%
Detroit	--	--	1.11%	--
Las Vegas	300.00	-0.07%	0.29%	-1.90%
Los Angeles	453.21	0.38%	0.59%	1.39%
Miami	451.98	0.32%	0.61%	2.27%
Minneapolis	253.44	0.70%	1.09%	1.89%
New York	352.23	1.03%	1.39%	4.79%
Phoenix	326.26	-0.10%	0.46%	-0.88%
Portland	334.30	0.27%	0.51%	-0.38%
San Diego	450.34	-0.14%	-0.17%	1.05%
San Francisco	371.01	-0.08%	0.84%	3.22%
Seattle	393.21	-0.47%	0.47%	-1.95%
Tampa	374.42	0.72%	0.66%	-1.19%
Washington	344.05	0.40%	0.75%	1.70%
Composite-10	373.49	0.52%	0.95%	2.94%
Composite-20	350.20	0.43%	0.89%	2.10%
U.S. National	336.66	0.37%	0.70%	1.52%

Sources: S&P Dow Jones Indices and Cotality
Data through June 2026

The same release, Table 2: the unadjusted June index LEVEL. The national unadjusted series ties to 336.66 here, directly. Detroit shows no June value at all, which is the release agreeing with our workbook about an absence.

The residual was explained, not tolerated. Comparing at full precision, five of twenty-two lines missed the printed value by 0.005 to 0.0074 percentage points — tiny, and outside the rounding of a two-decimal percent. Widening the tolerance would have turned them green and taught nobody anything. Running the same arithmetic on the previous month, which the same table publishes, showed no series missing twice and the signs random: noise, not a data difference. The cause is that S&P publishes levels to two decimals and computes its printed change from those. Rounding our two cells the same way reproduces the release **exactly** on 42 of 42 comparisons across both months — a stricter test than the one that was failing, not a looser one.

Every line in this set

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
ATXRSA	Case-Shiller HPI -- Atlanta, GA	Raw_Price!B7492	-0.12	-0.12	0.0	TIED
BOXRSA	Case-Shiller HPI -- Boston, MA	Raw_Price!B7596	0.11	0.11	0.0	TIED
CEXRSA	Case-Shiller HPI -- Cleveland, OH	Raw_Price!B7908	0.48	0.48	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
CHXRSA	Case-Shiller HPI -- Chicago, IL	Raw_Price!B7804	0.35	0.35	0.0	TIED
CRXRSA	Case-Shiller HPI -- Charlotte, NC	Raw_Price!B7700	-0.03	-0.03	0.0	TIED
CSUSHPINSA	S&P CoreLogic Case-Shiller U.S. National HPI (NSA)	Raw_Price!B212	336.66	336.66	0.0	TIED
CSUSHPISA	S&P CoreLogic Case-Shiller U.S. National HPI (SA)	Raw_Price!B316	0.13	0.13	0.0	TIED
DAXRSA	Case-Shiller HPI -- Dallas, TX	Raw_Price!B8012	-0.07	-0.07	0.0	TIED
DEXRSA	Case-Shiller HPI -- Detroit, MI	Raw_Price!B8220	0.15	0.15	0.0	TIED
DNXRSA	Case-Shiller HPI -- Denver, CO	Raw_Price!B8116	0.1	0.1	0.0	TIED
LVXRSA	Case-Shiller HPI -- Las Vegas, NV	Raw_Price!B8324	-0.5	-0.5	0.0	TIED
LXXRSA	Case-Shiller HPI -- Los Angeles, CA	Raw_Price!B8428	0.39	0.39	0.0	TIED
MIXRSA	Case-Shiller HPI -- Miami, FL	Raw_Price!B8532	-0.1	-0.1	0.0	TIED
MNXRSA	Case-Shiller HPI -- Minneapolis, MN	Raw_Price!B8636	0.15	0.15	0.0	TIED
NYXRSA	Case-Shiller HPI -- New York, NY	Raw_Price!B8740	0.72	0.72	0.0	TIED
PHXRSA	Case-Shiller HPI -- Phoenix, AZ	Raw_Price!B8844	-0.4	-0.4	0.0	TIED
POXRSA	Case-Shiller HPI -- Portland, OR	Raw_Price!B8948	0.11	0.11	0.0	TIED
SDXRSA	Case-Shiller HPI -- San Diego, CA	Raw_Price!B9052	-0.11	-0.11	0.0	TIED
SEXRSA	Case-Shiller HPI -- Seattle, WA	Raw_Price!B9260	-0.34	-0.34	0.0	TIED
SFXRSA	Case-Shiller HPI -- San Francisco, CA	Raw_Price!B9156	0.31	0.31	0.0	TIED
TPXRSA	Case-Shiller HPI -- Tampa, FL	Raw_Price!B9364	0.26	0.26	0.0	TIED
WDXRSA	Case-Shiller HPI -- Washington, DC	Raw_Price!B9468	0.41	0.41	0.0	TIED

And the workbook cell behind each one

	A	B	C
7490	ATXRSA	Case-Shiller HPI -- freq=monthly; units=index 2000-01=100; transform=yoy_p	
7491	date	value	
7492	2026-06-01	249.4971955	
7493	2026-05-01	249.802352	
7494	2026-04-01	249.8140597	

ATXRSA **TIED**

source: Table 3, row 'Atlanta', column 'SA' under 'June / May Change (%)'

	A	B	C
7594	BOXRSA	Case-Shiller HPI -- freq=monthly; units=index 2000-01=100; transform=yoy_p	
7595	date	value	
7596	2026-06-01	356.3176421	
7597	2026-05-01	355.9423328	
7598	2026-04-01	353.9919489	

BOXRSA **TIED**

source: Table 3, row 'Boston', column 'SA' under 'June / May Change (%)'

	A	B	C
7906	CEXRSA	Case-Shiller HPI -- freq=monthly; units=index 2000-01=100; transform=yoy_p	
7907	date	value	
7908	2026-06-01	207.4784072	
7909	2026-05-01	206.4854566	
7910	2026-04-01	206.7479917	

CEXRSA **TIED**

source: Table 3, row 'Cleveland', column 'SA' under 'June / May Change (%)'

	A	B	C
7802	CHXRSA	Case-Shiller HPI -- freq=monthly; units=index 2000-01=100; transform=yoy_p	
7803	date	value	
7804	2026-06-01	233.5382536	
7805	2026-05-01	232.718427	
7806	2026-04-01	231.4568922	

CHXRSA **TIED**

source: Table 3, row 'Chicago', column 'SA' under 'June / May Change (%)'

	A	B	C
7698	CRXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
7699	date	value	
7700	2026-06-01	285.440658	
7701	2026-05-01	285.531764	
7702	2026-04-01	286.2296163	

CRXRSA **TIED**

source: Table 3, row 'Charlotte', column 'SA' under 'June / May Change (%)'

	A	B	C
210	CSUSHPinsa	S&P CoreLogic Cē freq=monthly; units=index 2000-01=100; transform=index	
211	date	value	
212	2026-06-01	336.663	
213	2026-05-01	335.43	
214	2026-04-01	333.109	

CSUSHPinsa **TIED**

source: Table 2, row 'U.S. National', column 'June 2026 Level'

	A	B	C
314	CSUSHPISA	S&P CoreLogic Cē freq=monthly; units=index 2000-01=100; transform=index	
315	date	value	
316	2026-06-01	331.893	
317	2026-05-01	331.45	
318	2026-04-01	331.373	

CSUSHPISA **TIED**

source: Table 3, row 'U.S. National', column 'SA' under 'June / May Change (%)'

	A	B	C
8010	DAXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8011	date	value	
8012	2026-06-01	291.3526648	
8013	2026-05-01	291.5352974	
8014	2026-04-01	291.3635427	

DAXRSA **TIED**

source: Table 3, row 'Dallas', column 'SA' under 'June / May Change (%)'

	A	B	C
8218	DEXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8219	date	value	
8220	2026-05-01	201.4843573	
8221	2026-04-01	201.1846069	
8222	2026-03-01	200.1860873	

DEXRSA **TIED**

source: Table 3, row 'Detroit', column 'SA' under 'May / April Change (%)'

	A	B	C
8114	DNXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8115	date	value	
8116	2026-06-01	310.9058905	
8117	2026-05-01	310.5933009	
8118	2026-04-01	311.6294662	

DNXRSA **TIED**

source: Table 3, row 'Denver', column 'SA' under 'June / May Change (%)'

	A	B	C
8322	LVXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8323	date	value	
8324	2026-06-01	296.3915222	
8325	2026-05-01	297.8821857	
8326	2026-04-01	299.5302584	

LVXRSA **TIED**

source: Table 3, row 'Las Vegas', column 'SA' under 'June / May Change (%)'

	A	B	C
8426	LXXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8427	date	value	
8428	2026-06-01	445.1569615	
8429	2026-05-01	443.4393467	
8430	2026-04-01	442.2203898	

LXXRSA **TIED**

source: Table 3, row 'Los Angeles', column 'SA' under 'June / May Change (%)'

	A	B	C
8530	MIXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8531	date	value	
8532	2026-06-01	447.5187624	
8533	2026-05-01	447.9895686	
8534	2026-04-01	447.5918579	

MIXRSA **TIED**

source: Table 3, row 'Miami', column 'SA' under 'June / May Change (%)'

	A	B	C
8634	MNXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8635	date	value	
8636	2026-06-01	248.6208787	
8637	2026-05-01	248.2636697	
8638	2026-04-01	248.5076944	

MNXRSA **TIED**

source: Table 3, row 'Minneapolis', column 'SA' under 'June / May Change (%)'

	A	B	C
8738	NYXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8739	date	value	
8740	2026-06-01	347.570265	
8741	2026-05-01	345.094151	
8742	2026-04-01	343.3949445	

NYXRSA **TIED**

source: Table 3, row 'New York', column 'SA' under 'June / May Change (%)'

	A	B	C
8842	PHXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8843	date	value	
8844	2026-06-01	322.7436313	
8845	2026-05-01	324.0231787	
8846	2026-04-01	324.7478249	

PHXRSA **TIED**

source: Table 3, row 'Phoenix', column 'SA' under 'June / May Change (%)'

	A	B	C
8946	POXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
8947	date	value	
8948	2026-06-01	328.3243452	
8949	2026-05-01	327.9673386	
8950	2026-04-01	328.653117	

POXRSA **TIED**

source: Table 3, row 'Portland', column 'SA' under 'June / May Change (%)'

	A	B	C
9050	SDXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
9051	date	value	
9052	2026-06-01	440.1804477	
9053	2026-05-01	440.6674676	
9054	2026-04-01	443.0157142	

SDXRSA **TIED**

source: Table 3, row 'San Diego', column 'SA' under 'June / May Change (%)'

	A	B	C
9258	SEXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
9259	date	value	
9260	2026-06-01	382.5923449	
9261	2026-05-01	383.9102017	
9262	2026-04-01	384.9391027	

SEXRSA **TIED**

source: Table 3, row 'Seattle', column 'SA' under 'June / May Change (%)'

	A	B	C
9154	SFXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
9155	date	value	
9156	2026-06-01	361.3754276	
9157	2026-05-01	360.2795058	
9158	2026-04-01	359.5221505	

SFXRSA **TIED**

source: Table 3, row 'San Francisco', column 'SA' under 'June / May Change (%)'

	A	B	C
9362	TPXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
9363	date	value	
9364	2026-06-01	369.6112913	
9365	2026-05-01	368.6423446	
9366	2026-04-01	369.4624774	

TPXRSA **TIED**

source: Table 3, row 'Tampa', column 'SA' under 'June / May Change (%)'

	A	B	C
9466	WDXRSA	Case-Shiller HPI ~ freq=monthly; units=index 2000-01=100; transform=yoy_p	
9467	date	value	
9468	2026-06-01	338.5129719	
9469	2026-05-01	337.1394165	
9470	2026-04-01	336.6870769	

WDXRSA **TIED**

source: Table 3, row 'Washington', column 'SA' under 'June / May Change (%)'

Z.1 commercial property price indexes

Federal Reserve Board — **2 of 2 tied**

These two nearly went down as **COULD NOT**, and the obstacle was real but wrong. The Z.1 release ships a public CSV bundle; reading every member of it and collecting the series codes gives 6,107 codes, and neither of ours is among them. The Board's Financial Accounts Guide has a page for each code, but it prints the definition and no data, and its clipboard flow came back empty. Two dead ends, both accurately described. The third route works: the Board's Data Download Program publishes the entire Z.1 as one package, and that package carries both series. *It is not in the file I looked in* was never the same statement as *the Board does not publish it*.

One more thing had to be right before these could tie. FRED stamps a quarter at its first day and the Board stamps it at its last, so matching on the literal date reported “no observation” for a series that is plainly published. Same quarter, two conventions.

Every line in this set

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
B0GZ1FL075035403Q	Z.1 Multifamily Residential Real Estate Value, Level	Raw_Price!B524	329879	329879.0	0.0	TIED

Series	What it is	Workbook cell	Ours	Publisher	Diff	Verdict
BOGZ1FL075035503Q	Z.1 Commercial Real Estate Value (Nonfinancial Corporate), Level	Raw_Price!B420	314348	314348.0	0.0	TIED

And the workbook cell behind each one

	A	B	C
522	BOGZ1FL0750	Z.1 Multifamily R	<i>freq=quarterly; units=millions \$; transform=level; vintage=</i>
523	date	value	
524	2026-01-01	329879	
525	2025-10-01	327231	
526	2025-07-01	320920	

BOGZ1FL075035403Q **TIED**
source: series FL075035403.Q, observation 2026-03-31

	A	B	C
418	BOGZ1FL0750	Z.1 Commercial F	<i>freq=quarterly; units=millions \$; transform=level; vintage=</i>
419	date	value	
420	2026-01-01	314348	
421	2025-10-01	306187	
422	2025-07-01	307288	

BOGZ1FL075035503Q **TIED**
source: series FL075035503.Q, observation 2026-03-31

How to run the whole thing yourself

Real values, already filled in. Nothing here needs editing before it will run.

1. Rebuild the workbook, so the cells you check are the ones the code produces:

```
cd C:\Users\ajish\SATC-cs\credit-suite
$env:FRED_API_KEY = "<your key>"
python -m credit_suite.sources.fred.runner --live
```

2. Open `example-output\FRED_Credit_Risk_Dashboard.xlsm` and go to `Raw_Consumer`. Every series is a header row carrying its id and title, then date/value rows newest first.
3. Fetch the Board's own G.19 table and read the row for the newest month:

https://www.federalreserve.gov/releases/g19/hist/cc_hist_sa_levels.html

4. For the house price indexes, fetch FHFA's own quarterly files:

https://www.fhfa.gov/hpi/download/quarterly_datasets/hpi_at_state.csv
https://www.fhfa.gov/hpi/download/quarterly_datasets/hpi_at_metro.csv
https://www.fhfa.gov/hpi/download/quarterly_datasets/hpi_at_us_and_census.csv

5. For the charge-off and delinquency rates, the Board publishes six tables:

<https://www.federalreserve.gov/releases/chargeoff/chgallsa.htm>
<https://www.federalreserve.gov/releases/chargeoff/delallsa.htm>
<https://www.federalreserve.gov/releases/chargeoff/chgtop100sa.htm>
<https://www.federalreserve.gov/releases/chargeoff/deltop100sa.htm>
<https://www.federalreserve.gov/releases/chargeoff/chgothersa.htm>
<https://www.federalreserve.gov/releases/chargeoff/delothersa.htm>

6. For the survey, the Board's chart data and its Table 1:

```
https://www.federalreserve.gov/data/sloos/sloos-202607-chart-data.htm
https://www.federalreserve.gov/data/sloos/sloos-202607-table-1.htm
```

7. For the debt service ratios, use the CURRENT release, not the archive:

```
https://www.federalreserve.gov/releases/dsr/
```

8. For Z.1, the complete package rather than the release page's CSV bundle:

```
https://www.federalreserve.gov/datadownload/Output.aspx?rel=Z1&filetype=zip
```

9. For Case-Shiller, the monthly release. Tables 2 and 3 carry the levels and the adjusted changes:

```
https://www.spglobal.com/spdji/en/index-family/indicators/sp-cotality-case-shiller/
```

What this found

Four things, none of which any test in the suite could have caught, because every one of them left the numbers correct.

1 · A shipped workbook with a state missing from it

Nebraska's house price index was blank. FRED publishes it, with data through April 2026, and it fetched first time when asked again. The build had hit one `Internal Server Error`, recorded it honestly in its own status — `"errors": ["NESTHPI: Internal Server Error"]` — blanked the slot, and passed. The retry logic covered rate limits only, so a single 5xx out of 142 requests shipped a monitor with a state missing. **Fixed:** transient server errors, timeouts and dropped connections are now retried; a dead series id still fails on the first try, because that is how five retired series were caught earlier and retrying it would hide the next one.

2 · Two series wearing each other's description

`SUBLPDRCS` is commercial real estate **construction and land development** and was labelled nonfarm nonresidential. `SUBLPDRCSN` is **nonfarm nonresidential** and was labelled construction and land. Both values were right, on the dashboard and in the raw tabs, which is exactly why nothing noticed.

3 · A tightening indicator with its alert switched off

The costly one. `SUBLPDHMS` measures banks tightening standards on GSE-eligible mortgages. It was labelled a *demand* series, and the seed's helper gives demand series `alert_rule = "none"` — right for a demand series, and this is not one. So a real mortgage-tightening signal sat on the dashboard and could

never fire. **Fixed**, and that turns an alert on: it is a change in what the dashboard flags, not just in what it says.

4 • Two more labels that described a different series

`DRTSSP` is subprime **mortgage** standards, not consumer loans. `SUBLPDCILSLGNQ` is **large banks** tightening standards, not increasing spreads. Right value, wrong label, is invisible until somebody reads the label against the publisher's own definition.

5 • Four series that told the reader “billions” beside a figure in millions

Look again at the workbook shot in the worked example above. The metadata line reads `units=billions` \$ and the value under it is **5,166,907.71**. Five quadrillion dollars of consumer credit. The number is right — it ties to the cent against the Board's own table, which prints that figure in **millions**, and FRED publishes all four of these as “Millions of U.S. Dollars”. Only the declared unit was wrong, by a factor of a thousand, on the line a person reads.

Nothing computes with that field; it is displayed. That is precisely why no numeric test could see it, and why it took photographing the cell to notice. **Fixed**, with the declared unit now checked against the publisher's own. The shots in this document still say “billions”, deliberately: they are the evidence, and replacing them would erase the finding.

One thing deliberately left alone. The two Z.1 commercial-property series carry a unit I could not settle. Our config calls both “millions \$”. FRED labels one “Mil. of \$” and the other “%”. The Board's own Financial Accounts Guide describes both identically, as price indexes — and calls neither a dollar level. Three authorities, three answers, and the values themselves tie exactly against the Board's package. I could not establish which unit is right, so nothing was changed and it is written down here instead. Unknown is a third answer.

The guard that now holds it down. `tests/test_fred_labels.py` checks each label against the publisher's own definition: a series must carry the words its definition requires, must *not* carry a sibling's words (which is what catches a swap), and a tightening measure must not be wired as a demand one. It refuses if a survey series is added with no recorded definition, so the check cannot quietly stop covering things. Four mutations against it, all killed, on a 409-test baseline.

What I got wrong

- **I read the wrong debt-service page first.** The Board's `/releases/housedebt/` page is an archive, frozen at 2024:1 and last updated in September 2024, because the DSR moved to a credit-bureau methodology from the 2024:Q2 publication. Comparing a 2026 figure against it produced three apparent differences of 1.4 percentage points. Nothing was wrong with the data; I had picked a

discontinued series computed a different way. It is in this document as a photograph, above, because it is the exact failure the “same basis” check exists to stop.

- **I called five Case-Shiller lines DIFFERS before I understood them.** They were within 0.0074 of a percentage point. The right response was not a wider tolerance; it was finding out why, which took one extra month of the same published table and turned 22 near-misses into 22 exact matches.

- **Six of my first twelve source photographs were blank and reported “ok”.**

`window.scrollBy` produces an empty capture in headless Chrome, so every one of them came back as an identical 6,615-byte white rectangle. I only know because I opened them. Nothing in the pipeline would have told me.

- **My first Z.1 answer was going to be COULD NOT**, with an obstacle I could describe precisely and had not finished testing. It was two clicks from being wrong in a document.

What this does not prove

- **It proves the latest observation, not the whole history.** Each series was checked at its most recent period — 142 of 142 — and the Case-Shiller set additionally at the month before. The other 13,763 observations in the workbook were not individually compared to anything.
- **The twenty adjusted Case-Shiller metros are tied through a ratio, not a level.** S&P does not publish adjusted levels free. Two adjacent workbook cells reproduce S&P's published change exactly, which pins the month-on-month move and the relationship between those two cells. It would not catch a constant scale error applied to the whole series.
- **It proves the number, not the formula.** Nothing here checks the dashboard's z-scores, bands or alert logic. That was deliberate and it is the firm's own instruction: the mission was the base data.
- **The column maps rest on the tables' own headings.** Where a Federal Reserve table has eleven unlabelled numeric columns, the mapping was read off the header and then confirmed by the values agreeing across every series in that table. A mapping that is wrong in a way that still agrees on every series is not excluded, though it is hard to construct.
- **Nothing here was checked by a second person.**